

2.7 ▲ Derivatives of Exponential Functions

Now that we understand how derivatives interact with products and quotients, we are able to compute derivatives of

- polynomials,
- rational functions, and
- powers and roots of rational functions.

Notice that all of the above come from knowing¹⁷ the derivative of x^n and applying linearity of derivatives and the product rule.

There is still one more “rule” that we need to complete our toolbox and that is the chain rule. However before we get there, we will add a few functions to our list of things we can differentiate¹⁸. The first of these is the exponential function.

Let $a > 0$ and set $f(x) = a^x$ — this is what is known as an exponential function. Let’s see what happens when we try to compute the derivative of this function just using the definition of the derivative.

$$\begin{aligned}\frac{df}{dx} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{a^{x+h} - a^x}{h} \\ &= \lim_{h \rightarrow 0} a^x \cdot \frac{a^h - 1}{h} = a^x \cdot \lim_{h \rightarrow 0} \frac{a^h - 1}{h}\end{aligned}$$

Unfortunately we cannot complete this computation because we cannot evaluate the last limit directly. For the moment, let us assume this limit exists and name it

$$C(a) = \lim_{h \rightarrow 0} \frac{a^h - 1}{h}$$

It depends only on a and is completely independent of x . Using this notation (which we will quickly improve upon below), our desired derivative is now

$$\frac{d}{dx} a^x = C(a) \cdot a^x.$$

Thus the derivative of a^x is a^x multiplied by some constant — i.e. the function a^x is nearly unchanged by differentiating. If we can tune a so that $C(a) = 1$ then the derivative would just be the original function! This turns out to be very useful.

To try finding an a that obeys $C(a) = 1$, let us investigate how $C(a)$ changes with a . Unfortunately (though this fact is not at all obvious) there is no way to write $C(a)$ as a finite combination of any of the functions we have examined so far¹⁹. To get started, we’ll try to guess $C(a)$, for a few values of a , by plugging in some small values of h .

17 Differentiating powers and roots of functions is actually quite a bit easier once one knows the chain rule — which we will discuss soon.

18 One reason we add these functions is that they interact very nicely with the derivative. Another reason is that they turn up in many “real world” examples.

19 To a bit more be precise, we say that a number q is algebraic if we can write q as the zero of a polynomial with integer coefficients. When a is any positive algebraic number other than 1, $C(a)$ is not algebraic. A number that is not algebraic is called transcendental. The best known example of a transcendental number is π (which follows from the Lindemann-Weierstrass Theorem — way beyond the scope of this course).

Example 2.7.1

Let $a = 1$ then $C(1) = \lim_{h \rightarrow 0} \frac{1^h - 1}{h} = 0$. This is not surprising since $1^x = 1$ is constant, and so its derivative must be zero everywhere. Let $a = 2$ then $C(2) = \lim_{h \rightarrow 0} \frac{2^h - 1}{h}$. Setting h to smaller and smaller numbers gives

h	0.1	0.01	0.001	0.0001	0.00001	0.000001	0.0000001
$\frac{2^h - 1}{h}$	0.7177	0.6956	0.6934	0.6932	0.6931	0.6931	0.6931

Similarly when $a = 3$ we get

h	0.1	0.01	0.001	0.0001	0.00001	0.000001	0.0000001
$\frac{3^h - 1}{h}$	1.1612	1.1047	1.0992	1.0987	1.0986	1.0986	1.0986

and $a = 10$

h	0.1	0.01	0.001	0.0001	0.00001	0.000001	0.0000001
$\frac{10^h - 1}{h}$	2.5893	2.3293	2.3052	2.3028	2.3026	2.3026	2.3026

From this example it appears that $C(a)$ increases as we increase a , and that $C(a) = 1$ for some value of a between 2 and 3.

Example 2.7.1

We can learn a lot more about $C(a)$, and, in particular, confirm the guesses that we made in the last example, by making use of logarithms — this would be a good time for you to review them.

► Whirlwind Review of Logarithms

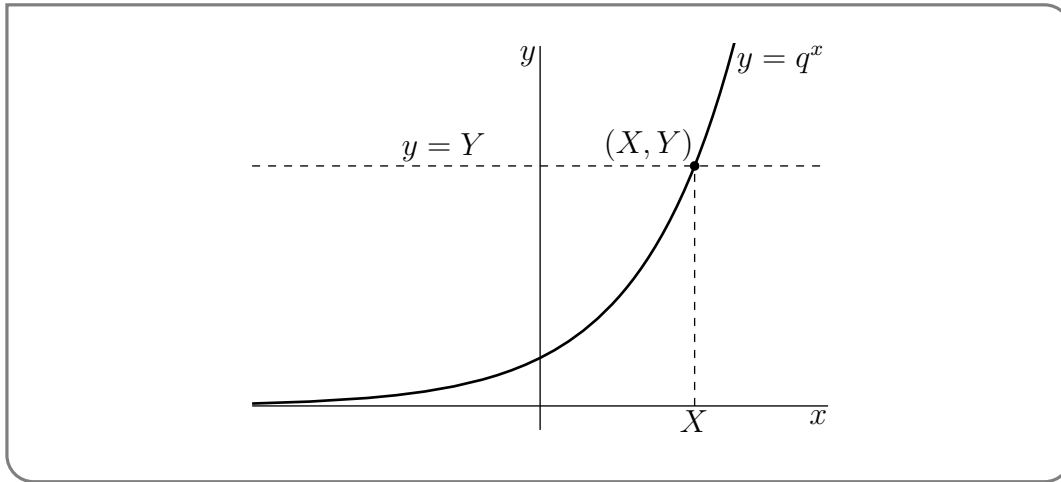
Before you read much further into this little review on logarithms, you should first go back and take a look at the review of inverse functions in Section [0.6](#).

►► Logarithmic Functions

We are about to define the “logarithm with base q ”. In principle, q is allowed to be any strictly positive real number, except $q = 1$. However we shall restrict our attention to $q > 1$, because, in practice, the only q ’s that are ever used are e (a number that we shall define in the next few pages), 10 and, if you are a computer scientist, 2. So, fix any $q > 1$ (if you like, pretend that $q = 10$). The function $f(x) = q^x$

- increases as x increases (for example if $x' > x$, then $10^{x'} = 10^x \cdot 10^{x'-x} > 10^x$ since $10^{x'-x} > 1$)
- obeys $\lim_{x \rightarrow -\infty} q^x = 0$ (for example 10^{-1000} is really small) and
- obeys $\lim_{x \rightarrow +\infty} q^x = +\infty$ (for example 10^{+1000} is really big).

Consequently, for any $0 < Y < \infty$, the horizontal straight line $y = Y$ crosses the graph of $y = f(x) = q^x$ at exactly one point, as illustrated in the figure below. The x -coordinate



of that intersection point, denoted X in the figure, is $\log_q(Y)$. So $\log_q(Y)$ is the power to which you have to raise q to get Y . It is the inverse function of $f(x) = q^x$. Of course we are free to rename the dummy variables X and Y . If, for example, we wish to graph our logarithm function, it is natural to rename $Y \rightarrow x$ and $X \rightarrow y$, giving

Definition 2.7.2.

Let $q > 1$. Then the logarithm with base q is defined²⁰ by

$$y = \log_q(x) \Leftrightarrow x = q^y$$

Obviously the power to which we have to raise q to get q^x is x , so we have both

$$\log_q(q^x) = x \qquad q^{\log_q(x)} = x$$

From the exponential properties

$$\begin{aligned} q^{\log_q(xy)} &= xy = q^{\log_q(x)} q^{\log_q(y)} = q^{\log_q(x) + \log_q(y)} \\ q^{\log_q(x/y)} &= x/y = q^{\log_q(x)} / q^{\log_q(y)} = q^{\log_q(x) - \log_q(y)} \\ q^{\log_q(x^r)} &= x^r = (q^{\log_q(x)})^r = q^{r \log_q(x)} \end{aligned}$$

²⁰ We can also define logarithms with base $0 < r < 1$ but doing so is not necessary. To see this, set $q = 1/r > 1$. Then it is reasonable to define $\log_r(x) = -\log_q(x)$ since

$$r^{\log_r(x)} = \left(\frac{1}{q}\right)^{\log_r(x)} = \left(\frac{1}{q}\right)^{-\log_q(x)} = q^{\log_q(x)} = x$$

as required.

we have

$$\begin{aligned}\log_q(xy) &= \log_q(x) + \log_q(y) \\ \log_q(x/y) &= \log_q(x) - \log_q(y) \\ \log_q(x^r) &= r \log_q(x)\end{aligned}$$

Can we convert from logarithms in one base to logarithms in another? For example, if our calculator computes logarithms base 10 for us (which it very likely does), can we also use it to compute a logarithm base q ? Yes, using

$$\log_q(x) = \frac{\log_{10} x}{\log_{10} q}$$

How did we get this? Well, let's start with a number x and suppose that we want to compute

$$y = \log_q x$$

We can rearrange this by exponentiating both sides

$$q^y = q^{\log_q x} = x$$

Now take log base 10 of both sides

$$\log_{10} q^y = \log_{10} x$$

But recall that $\log_q(x^r) = r \log_q(x)$, so

$$\begin{aligned}y \log_{10} q &= \log_{10} x \\ y &= \frac{\log_{10} x}{\log_{10} q}\end{aligned}$$

► Back to that Limit

Recall that we are trying to choose a so that

$$\lim_{h \rightarrow 0} \frac{a^h - 1}{h} = C(a) = 1.$$

We can estimate the correct value of a by using our numerical estimate of $C(10)$ above. The way to do this is to first rewrite $C(a)$ in terms of logarithms.

$$a = 10^{\log_{10} a} \quad \text{and so} \quad a^h = 10^{h \log_{10} a}.$$

Using this we rewrite $C(a)$ as

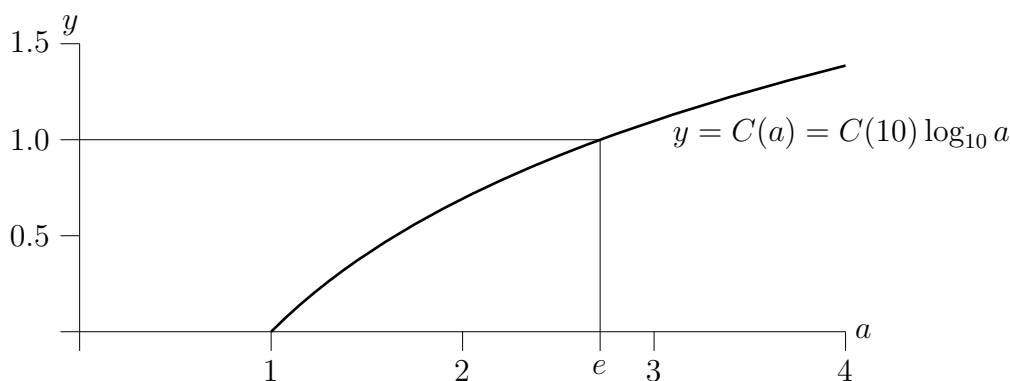
$$C(a) = \lim_{h \rightarrow 0} \frac{1}{h} \left(10^{h \log_{10} a} - 1 \right)$$

Now set $H = h \log_{10}(a)$, and notice that as $h \rightarrow 0$ we also have $H \rightarrow 0$

$$\begin{aligned} &= \lim_{H \rightarrow 0} \frac{\log_{10} a}{H} (10^H - 1) \\ &= \log_{10} a \cdot \lim_{H \rightarrow 0} \frac{10^H - 1}{H} \\ &= \log_{10} a \cdot C(10). \end{aligned}$$

Below is a sketch of $C(a)$ against a .

Figure 2.7.1.



Remember that we are trying to find an a with $C(a) = 1$. We can do so by recognising that $C(a) = C(10) (\log_{10} a)$ has the following properties.

- When $a = 1$, $\log_{10}(a) = \log_{10} 1 = 0$ so that $C(a) = C(10) \log_{10}(a) = 0$. Of course, we should have expected this, because when $a = 1$ we have $a^x = 1^x = 1$ which is just the constant function and $\frac{d}{dx} 1 = 0$.
- $\log_{10} a$ increases as a increases, and hence $C(a) = C(10) \log_{10} a$ increases as a increases.
- $\log_{10} a$ tends to $+\infty$ as $a \rightarrow \infty$, and hence $C(a)$ tends to $+\infty$ as $a \rightarrow \infty$.

Hence the graph of $C(a)$ passes through $(1, 0)$, is always increasing as a increases and goes off to $+\infty$ as a goes off to $+\infty$. See Figure 2.7.1. Consequently there is exactly one value of a for which $C(a) = 1$.

The value of a for which $C(a) = 1$ is given the name e . It is called Euler's constant²².

21 We are applying the Intermediate Value Theorem here, but we have neglected to verify the hypothesis that $\log_{10}(a)$ is a continuous function. Please forgive us — we could do this if we really had to, but it would make a big mess without adding much understanding, if we were to do so here in the text. Better to just trust us on this.

22 Unfortunately there is another Euler's constant, γ , which is more properly called the Euler–Mascheroni constant. Anyway like many mathematical discoveries, e was first found by someone else — Napier used the constant e in order to compute logarithms but only implicitly. Bernoulli was probably the first to approximate it when examining continuous compound interest. It first appeared explicitly in work of Leibniz, though he denoted it b . It was Euler, though, who established the notation we now use and who showed how important the constant is to mathematics.

In Example 2.7.1, we estimated $C(10) \approx 2.3026$. So if we assume $C(a) = 1$ then the above equation becomes

$$\begin{aligned} 2.3026 \cdot \log_{10} a &\approx 1 \\ \log_{10} a &\approx \frac{1}{2.3026} \approx 0.4343 \\ a &\approx 10^{0.4343} \approx 2.7813 \end{aligned}$$

This gives us the estimate $a \approx 2.7813$ which is not too bad. In fact 23

Equation 2.7.3 (Euler's constant).

$$\begin{aligned} e &= 2.7182818284590452354 \dots \\ &= 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots \end{aligned}$$

We will be able to explain this last formula once we develop Taylor polynomials later in the course.

To summarise

Theorem 2.7.4.

The constant e is the unique real number that satisfies

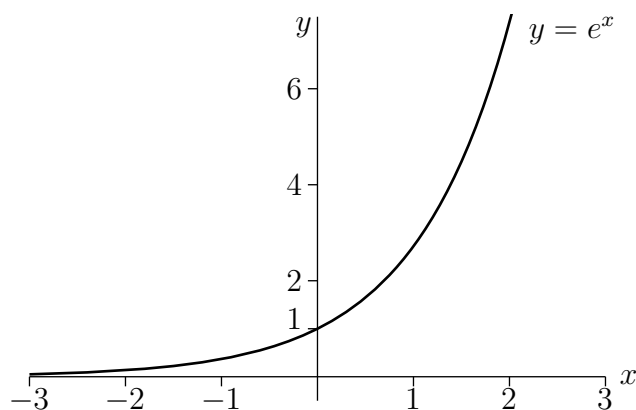
$$\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$$

Further,

$$\frac{d}{dx}(e^x) = e^x$$

We plot e^x in the graph below

23 Recall n factorial, written $n!$ is the product $n \times (n-1) \times (n-2) \times \dots \times 2 \times 1$.

Figure 2.7.2.

And just a reminder of some of its²⁴ properties...

1. $e^0 = 1$
2. $e^{x+y} = e^x e^y$
3. $e^{-x} = \frac{1}{e^x}$
4. $(e^x)^y = e^{xy}$
5. $\lim_{x \rightarrow \infty} e^x = \infty, \lim_{x \rightarrow -\infty} e^x = 0$

Now consider again the problem of differentiating a^x . We saw above that

$$\frac{d}{dx} a^x = C(a) \cdot a^x \quad \text{and } C(a) = C(10) \cdot \log_{10} a \quad \text{which gives } \frac{d}{dx} a^x = C(10) \cdot \log_{10} a \cdot a^x$$

We can eliminate the $C(10)$ term with a little care. Since we know that $\frac{d}{dx} e^x = e^x$, we have $C(e) = 1$. This allows us to express

$$1 = C(e) = C(10) \cdot \log_{10} e \quad \text{and so}$$

$$C(10) = \frac{1}{\log_{10} e}$$

Putting things back together gives

$$\begin{aligned} \frac{d}{dx} a^x &= \frac{\log_{10} a}{\log_{10} e} \cdot a^x \\ &= \log_e a \cdot a^x. \end{aligned}$$

24 "The function e^x is of course the special case of the function a^x with $a = e$. So it inherits all the usual algebraic properties of a^x ."

There is more than one way to get to this result. For example, let $f(x) = a^x$, then

$$\begin{aligned}\log_e f(x) &= x \log_e a \\ f(x) &= e^{x \log_e a}\end{aligned}$$

So if we write $g(x) = e^x$ then we are really attempting to differentiate the function

$$\frac{df}{dx} = \frac{d}{dx} g(x \cdot \log_e a).$$

In order to compute this derivative we need to know how to differentiate

$$\frac{d}{dx} g(qx)$$

where q is a constant. We'll hold off on learning this for the moment until we have introduced the chain rule (see Section 2.9 and in particular Corollary 2.9.9). Similarly we'd like to know how to differentiate logarithms — again this has to wait until we have learned the chain rule.

Notice that the derivatives

$$\frac{d}{dx} x^n = nx^{n-1} \quad \text{and} \quad \frac{d}{dx} e^x = e^x$$

are either nearly unchanged or actually unchanged by differentiating. It turns out that some of the trigonometric functions also have this property of being “nearly unchanged” by differentiation. That brings us to the next section.

2.8 ▲ Derivatives of Trigonometric Functions

We are now going to compute the derivatives of the various trigonometric functions, $\sin x$, $\cos x$ and so on. The computations are more involved than the others that we have done so far and will take several steps. Fortunately, the final answers will be very simple.

Observe that we only need to work out the derivatives of $\sin x$ and $\cos x$, since the other trigonometric functions are really just quotients of these two functions. Recall:

$$\tan x = \frac{\sin x}{\cos x} \quad \cot x = \frac{\cos x}{\sin x} \quad \csc x = \frac{1}{\sin x} \quad \sec x = \frac{1}{\cos x}.$$

The first steps towards computing the derivatives of $\sin x$, $\cos x$ is to find their derivatives at $x = 0$. The derivatives at general points x will follow quickly from these, using trig identities. It is important to note that we must measure angles in radians²⁵, rather than degrees, in what follows. Indeed — unless explicitly stated otherwise, any number that is put into a trigonometric function is measured in radians.

25 In science, radians is the standard unit for measuring angles. While you may be more familiar with degrees, radians should be used in any computation involving calculus. Using degrees will cause errors. Thankfully it is easy to translate between these two measures since $360^\circ = 2\pi$ radians. See Appendix B.2.1.

►► **These Proofs are Optional, the Results are Not.**

While we expect you to read and follow these proofs, we do not expect you to be able to reproduce them. You will be required to know the results, in particular Theorem [2.8.5](#) below.

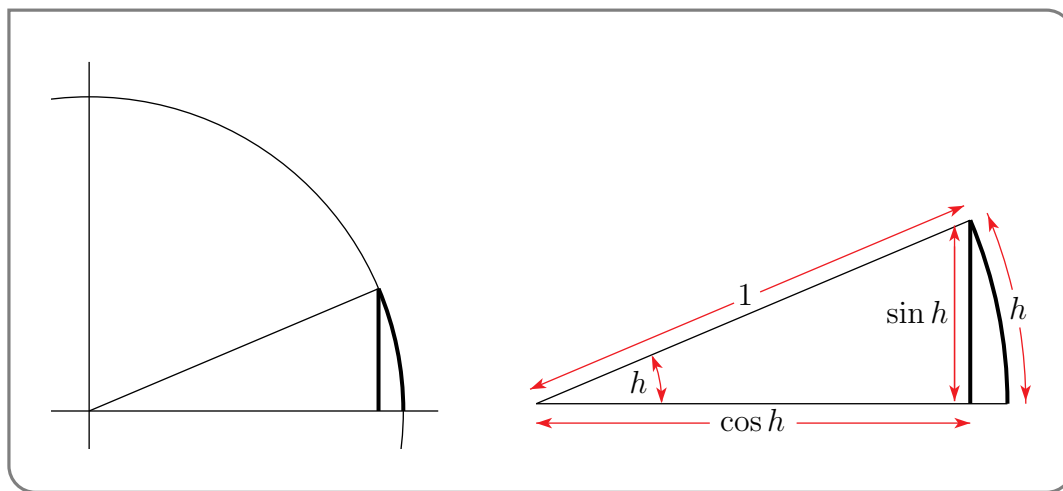
►► **Step 1:** $\frac{d}{dx}\{\sin x\}\big|_{x=0}$

By definition, the derivative of $\sin x$ evaluated at $x = 0$ is

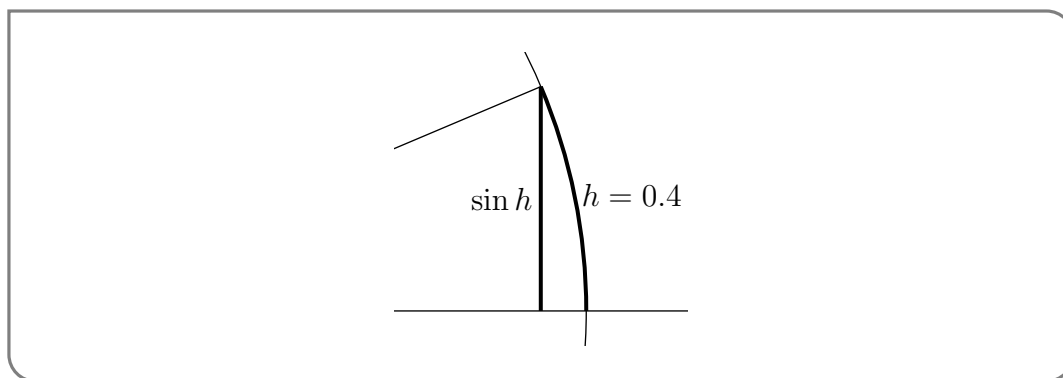
$$\frac{d}{dx}\{\sin x\}\big|_{x=0} = \lim_{h \rightarrow 0} \frac{\sin h - \sin 0}{h} = \lim_{h \rightarrow 0} \frac{\sin h}{h}$$

We will prove this limit by use of the squeeze theorem (Theorem [1.4.17](#)). To get there we will first need to do some geometry. But first we will build some intuition.

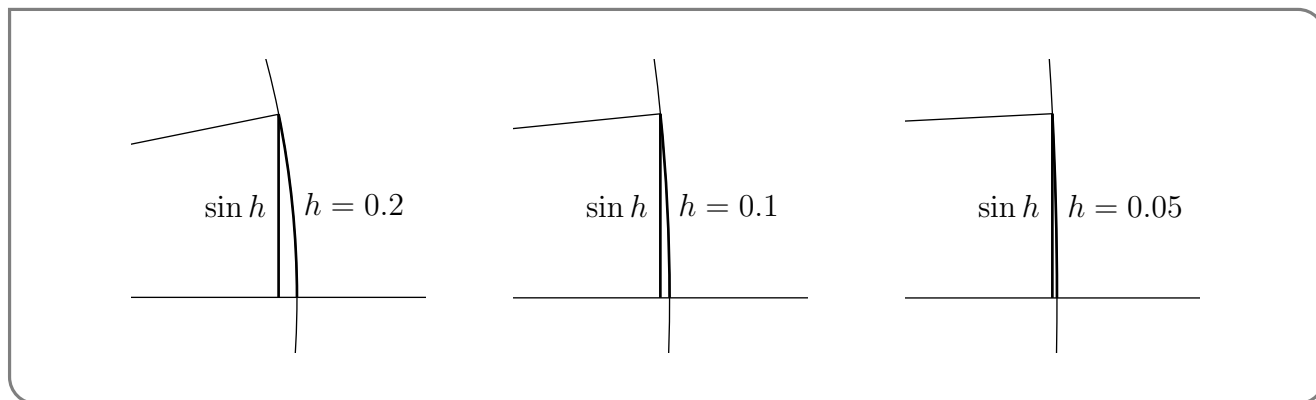
The figure below contains part of a circle of radius 1. Recall that an arc of length h on such a circle subtends an angle of h **radians** at the centre of the circle. So the darkened arc in the figure has length h and the darkened vertical line in the figure has length $\sin h$. We must determine what happens to the ratio of the lengths of the darkened vertical line and darkened arc as h tends to zero.



Here is a magnified version of the part of the above figure that contains the darkened arc and vertical line.



This particular figure has been drawn with $h = .4$ radians. Here are three more such blow ups. In each successive figure, the value of h is smaller. To make the figures clearer, the degree of magnification was increased each time h was decreased.



As we make h smaller and smaller and look at the figure with ever increasing magnification, the arc of length h and vertical line of length $\sin h$ look more and more alike. We would guess from this that

$$\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$$

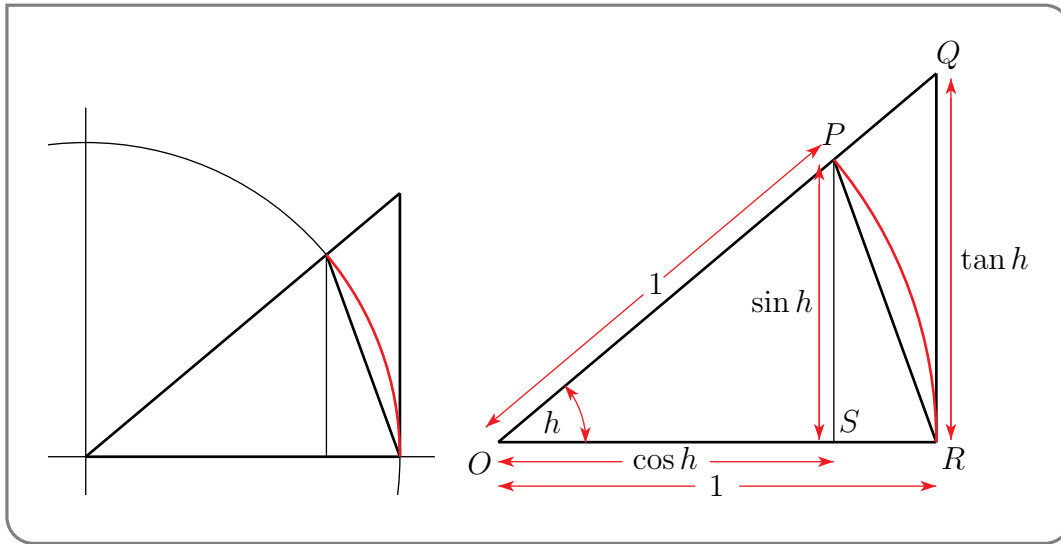
The following tables of values

h	$\sin h$	$\frac{\sin h}{h}$
0.4	.3894	.9735
0.2	.1987	.9934
0.1	.09983	.9983
0.05	.049979	.99958
0.01	.00999983	.999983
0.001	.0099999983	.9999983

h	$\sin h$	$\frac{\sin h}{h}$
-0.4	-.3894	.9735
-0.2	-.1987	.9934
-0.1	-.09983	.9983
-0.05	-.049979	.99958
-0.01	-.00999983	.999983
-0.001	-.0099999983	.9999983

suggests the same guess. Here is an argument that shows that the guess really is correct.

►►► **Proof that** $\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$:



The circle in the figure above has radius 1. Hence

$$\begin{aligned} |OP| &= |OR| = 1 & |PS| &= \sin h \\ |OS| &= \cos h & |QR| &= \tan h \end{aligned}$$

Now we can use a few geometric facts about this figure to establish both an upper bound and a lower bound on $\frac{\sin h}{h}$ with both the upper and lower bounds tending to 1 as h tends to 0. So the squeeze theorem will tell us that $\frac{\sin h}{h}$ also tends to 1 as h tends to 0.

- The triangle OPR has base 1 and height $\sin h$, and hence

$$\text{area of } \triangle OPR = \frac{1}{2} \times 1 \times \sin h = \frac{\sin h}{2}.$$

- The triangle OQR has base 1 and height $\tan h$, and hence

$$\text{area of } \triangle OQR = \frac{1}{2} \times 1 \times \tan h = \frac{\tan h}{2}.$$

- The “piece of pie” OPR cut out of the circle is the fraction $\frac{h}{2\pi}$ of the whole circle (since the angle at the corner of the piece of pie is h radians and the angle for the whole circle is 2π radians). Since the circle has radius 1 we have

$$\text{area of pie } OPR = \frac{h}{2\pi} \cdot (\text{area of circle}) = \frac{h}{2\pi} \pi \cdot 1^2 = \frac{h}{2}$$

Now the triangle OPR is contained inside the piece of pie OPR . and so the area of the triangle is smaller than the area of the piece of pie. Similarly, the piece of pie OPR is contained inside the triangle OQR . Thus we have

$$\text{area of triangle } OPR \leq \text{area of pie } OPR \leq \text{area of triangle } OQR$$

Substituting in the areas we worked out gives

$$\frac{\sin h}{2} \leq \frac{h}{2} \leq \frac{\tan h}{2}$$

which cleans up to give

$$\sin h \leq h \leq \frac{\sin h}{\cos h}$$

We rewrite these two inequalities so that $\frac{\sin h}{h}$ appears in both.

- Since $\sin h \leq h$, we have that $\frac{\sin h}{h} \leq 1$.
- Since $h \leq \frac{\sin h}{\cos h}$ we have that $\cos h \leq \frac{\sin h}{h}$.

Thus we arrive at the “squeezable” inequality

$$\cos h \leq \frac{\sin h}{h} \leq 1$$

We know²⁶ that

$$\lim_{h \rightarrow 0} \cos h = 1.$$

Since $\frac{\sin h}{h}$ is sandwiched between $\cos h$ and 1, we can apply the squeeze theorem for limits (Theorem 1.4.17) to deduce the following lemma:

Lemma 2.8.1.

$$\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1.$$

Since this argument took a bit of work, perhaps we should remind ourselves why we needed it in the first place. We were computing

$$\begin{aligned} \frac{d}{dx} \{\sin x\} \Big|_{x=0} &= \lim_{h \rightarrow 0} \frac{\sin h - \sin 0}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sin h}{h} && \text{(This is why!)} \\ &= 1 \end{aligned}$$

This concludes Step 1. We now know that $\frac{d}{dx} \sin x \Big|_{x=0} = 1$. The remaining steps are easier.

26 Again, refresh your memory by looking up Appendix A.5.

►► **Step 2:** $\left. \frac{d}{dx} \{\cos x\} \right|_{x=0}$

By definition, the derivative of $\cos x$ evaluated at $x = 0$ is

$$\lim_{h \rightarrow 0} \frac{\cos h - \cos 0}{h} = \lim_{h \rightarrow 0} \frac{\cos h - 1}{h}$$

Fortunately we don't have to wade through geometry like we did for the previous step. Instead we can recycle our work and massage the above limit to rewrite it in terms of expressions involving $\frac{\sin h}{h}$. Thanks to Lemma 2.8.1 the work is then easy.

We'll show you two ways to proceed — one uses a method similar to “multiplying by the conjugate” that we have already used a few times (see Example 1.4.16 and 2.2.9), while the other uses a nice trick involving the double-angle formula²⁷.

►►► **Method 1 — Multiply by the “Conjugate”**

Start by multiplying the expression inside the limit by 1, written as $\frac{\cos h + 1}{\cos h + 1}$:

$$\begin{aligned} \frac{\cos h - 1}{h} &= \frac{\cos h - 1}{h} \cdot \frac{\cos h + 1}{\cos h + 1} \\ &= \frac{\cos^2 h - 1}{h(1 + \cos h)} && \text{(since } (a - b)(a + b) = a^2 - b^2 \text{)} \\ &= -\frac{\sin^2 h}{h(1 + \cos h)} && \text{(since } \sin^2 h + \cos^2 h = 1 \text{)} \\ &= -\frac{\sin h}{h} \cdot \frac{\sin h}{1 + \cos h} \end{aligned}$$

Now we can take the limit as $h \rightarrow 0$ via Lemma 2.8.1

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} &= \lim_{h \rightarrow 0} \left(\frac{-\sin h}{h} \cdot \frac{\sin h}{1 + \cos h} \right) \\ &= -\lim_{h \rightarrow 0} \left(\frac{\sin h}{h} \right) \cdot \lim_{h \rightarrow 0} \left(\frac{\sin h}{1 + \cos h} \right) \\ &= -1 \cdot \frac{0}{2} \\ &= 0 \end{aligned}$$

►►► **Method 2 — via the Double Angle Formula**

The other way involves the double angle formula²⁸

$$\cos 2\theta = 1 - 2\sin^2(\theta) \quad \text{or} \quad \cos 2\theta - 1 = -2\sin^2(\theta)$$

²⁷ See Appendix A.14 if you have forgotten. You should also recall that $\sin^2 \theta + \cos^2 \theta = 1$. Sorry for nagging.

²⁸ We hope you looked this up in in Appendix A.14. Nag.

Setting $\theta = h/2$, we have

$$\frac{\cos h - 1}{h} = \frac{-2\left(\sin \frac{h}{2}\right)^2}{h}$$

Now this begins to look like $\frac{\sin h}{h}$, except that inside the $\sin(\cdot)$ we have $h/2$. So, setting $\theta = h/2$,

$$\begin{aligned}\frac{\cos h - 1}{h} &= -\frac{\sin^2 \theta}{\theta} = -\theta \cdot \frac{\sin^2 \theta}{\theta^2} \\ &= -\theta \cdot \frac{\sin \theta}{\theta} \cdot \frac{\sin \theta}{\theta}\end{aligned}$$

When we take the limit as $h \rightarrow 0$, we are also taking the limit as $\theta = h/2 \rightarrow 0$, and so

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{\cos h - 1}{h} &= \lim_{\theta \rightarrow 0} \left[-\theta \cdot \frac{\sin \theta}{\theta} \cdot \frac{\sin \theta}{\theta} \right] \\ &= \lim_{\theta \rightarrow 0} [-\theta] \cdot \lim_{\theta \rightarrow 0} \left[\frac{\sin \theta}{\theta} \right] \cdot \lim_{\theta \rightarrow 0} \left[\frac{\sin \theta}{\theta} \right] \\ &= 0 \cdot 1 \cdot 1 \\ &= 0\end{aligned}$$

where we have used the fact that $\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$ and that the limit of a product is the product of limits (i.e. Lemma 2.8.1 and Theorem 1.4.2).

Thus we have now produced two proofs of the following lemma:

Lemma 2.8.2.

$$\lim_{h \rightarrow 0} \frac{\cos h - 1}{h} = 0$$

Again, there has been a bit of work to get to here, so we should remind ourselves why we needed it. We were computing

$$\begin{aligned}\frac{d}{dx} \{\cos x\} \Big|_{x=0} &= \lim_{h \rightarrow 0} \frac{\cos h - \cos 0}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} \\ &= 0\end{aligned}$$

Armed with these results we can now build up the derivatives of sine and cosine.

►► **Step 3:** $\frac{d}{dx}\{\sin x\}$ and $\frac{d}{dx}\{\cos x\}$ for General x

To proceed to the general derivatives of $\sin x$ and $\cos x$ we are going to use the above two results and a couple of trig identities. Remember the addition formulae²⁹

$$\sin(a + b) = \sin(a) \cos(b) + \cos(a) \sin(b)$$

$$\cos(a + b) = \cos(a) \cos(b) - \sin(a) \sin(b)$$

To compute the derivative of $\sin(x)$ we just start from the definition of the derivative:

$$\begin{aligned} \frac{d}{dx} \sin x &= \lim_{h \rightarrow 0} \frac{\sin(x + h) - \sin x}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} \\ &= \lim_{h \rightarrow 0} \left[\sin x \frac{\cos h - 1}{h} + \cos x \frac{\sin h - 0}{h} \right] \\ &= \sin x \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} + \cos x \lim_{h \rightarrow 0} \frac{\sin h - 0}{h} \\ &= \sin x \underbrace{\left[\frac{d}{dx} \cos x \right]_{x=0}}_{=0} + \cos x \underbrace{\left[\frac{d}{dx} \sin x \right]_{x=0}}_{=1} \\ &= \cos x \end{aligned}$$

The computation of the derivative of $\cos x$ is very similar.

$$\begin{aligned} \frac{d}{dx} \cos x &= \lim_{h \rightarrow 0} \frac{\cos(x + h) - \cos x}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cos x \cos h - \sin x \sin h - \cos x}{h} \\ &= \lim_{h \rightarrow 0} \left[\cos x \frac{\cos h - 1}{h} - \sin x \frac{\sin h - 0}{h} \right] \\ &= \cos x \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} - \sin x \lim_{h \rightarrow 0} \frac{\sin h - 0}{h} \\ &= \cos x \underbrace{\left[\frac{d}{dx} \cos x \right]_{x=0}}_{=0} - \sin x \underbrace{\left[\frac{d}{dx} \sin x \right]_{x=0}}_{=1} \\ &= -\sin x \end{aligned}$$

We have now found the derivatives of both $\sin x$ and $\cos x$, *provided x is measured in radians*.

29 You really should. Look this up in Appendix [A.8](#) if you have forgotten.

Lemma 2.8.3.

$$\frac{d}{dx} \sin x = \cos x$$

$$\frac{d}{dx} \cos x = -\sin x$$

The above formulas hold provided x is measured in radians.

These formulae are pretty easy to remember — applying $\frac{d}{dx}$ to $\sin x$ and $\cos x$ just exchanges $\sin x$ and $\cos x$, except for the minus sign³⁰ in the derivative of $\cos x$.

Remark 2.8.4 (Optional — Another derivation of $\frac{d}{dx} \cos x = -\sin x$). We remark that, once one knows that $\frac{d}{dx} \sin x = \cos x$, it is easy to use it and the trig identity $\cos(x) = \sin(\frac{\pi}{2} - x)$ to derive $\frac{d}{dx} \cos x = -\sin x$. Here is how³¹.

$$\begin{aligned} \frac{d}{dx} \cos x &= \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h} = \lim_{h \rightarrow 0} \frac{\sin(\frac{\pi}{2} - x - h) - \sin(\frac{\pi}{2} - x)}{h} \\ &= - \lim_{h' \rightarrow 0} \frac{\sin(x' + h') - \sin(x')}{h'} \quad \text{with } x' = \frac{\pi}{2} - x, h' = -h \\ &= - \frac{d}{dx'} \sin x' \Big|_{x' = \frac{\pi}{2} - x} = -\cos(\frac{\pi}{2} - x) \\ &= -\sin x \end{aligned}$$

Note that, if x is measured in degrees, then the formulas of Lemma 2.8.3 are wrong. There are similar formulas, but we need the chain rule to build them — that is the subject of the next section. But first we should find the derivatives of the other trig functions.

► Step 4: the Remaining Trigonometric Functions

It is now an easy matter to get the derivatives of the remaining trigonometric functions using basic trig identities and the quotient rule. Remember³² that

$$\begin{aligned} \tan x &= \frac{\sin x}{\cos x} \\ \csc x &= \frac{1}{\sin x} \end{aligned}$$

$$\begin{aligned} \cot x &= \frac{\cos x}{\sin x} = \frac{1}{\tan x} \\ \sec x &= \frac{1}{\cos x} \end{aligned}$$

30 There is a bad pun somewhere in here about sine errors and sign errors.

31 We thank Serban Raianu for suggesting that we include this.

32 You really should. If you do not then take a quick look at the appropriate appendix.

So, by the quotient rule,

$$\begin{aligned}\frac{d}{dx} \tan x &= \frac{d}{dx} \frac{\sin x}{\cos x} = \frac{\overbrace{\left(\frac{d}{dx} \sin x\right)}^{\cos x} \cos x - \sin x \overbrace{\left(\frac{d}{dx} \cos x\right)}^{-\sin x}}{\cos^2 x} = \sec^2 x \\ \frac{d}{dx} \csc x &= \frac{d}{dx} \frac{1}{\sin x} = -\frac{\overbrace{\left(\frac{d}{dx} \sin x\right)}^{\cos x}}{\sin^2 x} = -\csc x \cot x \\ \frac{d}{dx} \sec x &= \frac{d}{dx} \frac{1}{\cos x} = -\frac{\overbrace{\left(\frac{d}{dx} \cos x\right)}^{-\sin x}}{\cos^2 x} = \sec x \tan x \\ \frac{d}{dx} \cot x &= \frac{d}{dx} \frac{\cos x}{\sin x} = \frac{\overbrace{\left(\frac{d}{dx} \cos x\right)}^{-\sin x} \sin x - \cos x \overbrace{\left(\frac{d}{dx} \sin x\right)}^{\cos x}}{\sin^2 x} = -\csc^2 x\end{aligned}$$

► Summary

To summarise all this work, we can write this up as a theorem:

Theorem 2.8.5 (Derivatives of trigonometric functions).

The derivatives of $\sin x$ and $\cos x$ are

$$\frac{d}{dx} \sin x = \cos x$$

$$\frac{d}{dx} \cos x = -\sin x$$

Consequently the derivatives of the other trigonometric functions are

$$\frac{d}{dx} \tan x = \sec^2 x$$

$$\frac{d}{dx} \cot x = -\csc^2 x$$

$$\frac{d}{dx} \csc x = -\csc x \cot x$$

$$\frac{d}{dx} \sec x = \sec x \tan x$$

Of these 6 derivatives you should really memorise those of sine, cosine and tangent. We certainly expect you to be able to work out those of cotangent, cosecant and secant.

2.9 ▲ One More Tool – the Chain Rule

We have built up most of the tools that we need to express derivatives of complicated functions in terms of derivatives of simpler known functions. We started by learning how to evaluate

- derivatives of sums, products and quotients

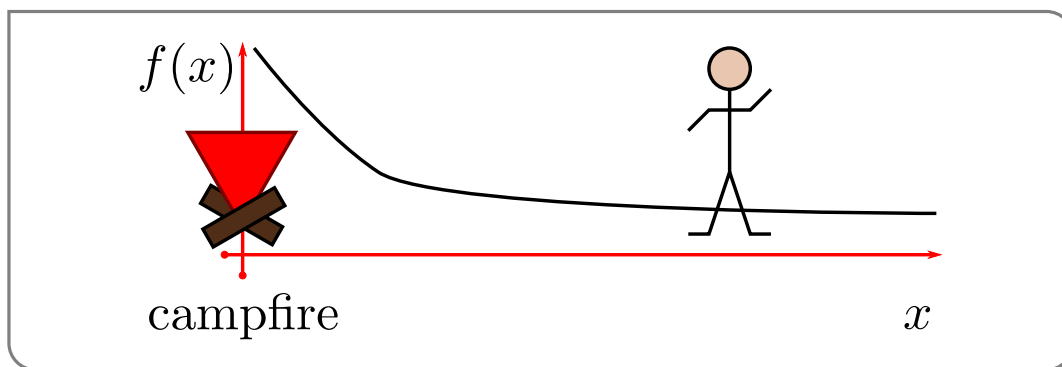
- derivatives of constants and monomials

These tools allow us to compute derivatives of polynomials and rational functions. In the previous sections, we added exponential and trigonometric functions to our list. The final tool we add is called the chain rule. It tells us how to take the derivative of a composition of two functions. That is if we know $f(x)$ and $g(x)$ and their derivatives, then the chain rule tells us the derivative of $f(g(x))$.

Before we get to the statement of the rule, let us look at an example showing how such a composition might arise (in the “real-world”).

Example 2.9.1

You are out in the woods after a long day of mathematics and are walking towards your camp fire on a beautiful still night. The heat from the fire means that the air temperature depends on your position. Let your position at time t be $x(t)$. The temperature of the air at position x is $f(x)$. What instantaneous rate of change of temperature do you feel at time t ?



- Because your position at time t is $x = x(t)$, the temperature you feel at time t is $F(t) = f(x(t))$.
- The instantaneous rate of change of temperature that you feel is $F'(t)$. We have a complicated function, $F(t)$, constructed by composing two simpler functions, $x(t)$ and $f(x)$.
- We wish to compute the derivative, $F'(t) = \frac{d}{dt}f(x(t))$, of the complicated function $F(t)$ in terms of the derivatives, $x'(t)$ and $f'(x)$, of the two simple functions. This is exactly what the chain rule does.

Example 2.9.1

► Statement of the Chain Rule

Theorem 2.9.2 (The chain rule — version 1).

Let $a \in \mathbb{R}$ and let $g(x)$ be a function that is differentiable at $x = a$. Now let $f(u)$ be a function that is differentiable at $u = g(a)$. Then the function $F(x) = f(g(x))$ is differentiable at $x = a$ and

$$F'(a) = f'(g(a)) g'(a)$$

Here, as was the case earlier in this chapter, we have been very careful to give the point at which the derivative is evaluated a special name (i.e. a). But of course this evaluation point can really be any point (where the derivative is defined). So it is very common to just call the evaluation point “ x ” rather than give it a special name like “ a ”, like this:

Theorem 2.9.3 (The chain rule — version 2).

Let f and g be differentiable functions then

$$\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$$

Notice that when we form the composition $f(g(x))$ there is an “outside” function (namely $f(x)$) and an “inside” function (namely $g(x)$). The chain rule tells us that when we differentiate a composition that we have to differentiate the outside and then multiply by the derivative of the inside.

$$\frac{d}{dx}f(g(x)) = \underbrace{f'(g(x))}_{\text{diff outside}} \cdot \underbrace{g'(x)}_{\text{diff inside}}$$

Here is another statement of the chain rule which makes this idea more explicit.

Theorem 2.9.4 (The chain rule — version 3).

Let $y = f(u)$ and $u = g(x)$ be differentiable functions, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

This particular form is easy to remember because it looks like we can just “cancel” the du between the two terms.

$$\frac{dy}{dx} = \frac{dy}{\cancel{du}} \cdot \frac{\cancel{du}}{dx}$$

Of course, du is not, by itself, a number or variable³³ that can be cancelled. But this is still a good memory aid.

The hardest part about applying the chain rule is recognising when the function you are trying to differentiate is really the composition of two simpler functions. This takes a little practice. We can warm up with a couple of simple examples.

Example 2.9.5

Let $f(u) = u^5$ and $g(x) = \sin(x)$. Then set $F(x) = f(g(x)) = (\sin(x))^5$. To find the derivative of $F(x)$ we can simply apply the chain rule — the pieces of the composition have been laid out for us. Here they are.

$$\begin{aligned} f(u) &= u^5 & f'(u) &= 5u^4 \\ g(x) &= \sin(x) & g'(x) &= \cos x \end{aligned}$$

We now just put them together as the chain rule tells us

$$\begin{aligned} \frac{dF}{dx} &= f'(g(x)) \cdot g'(x) \\ &= 5(g(x))^4 \cdot \cos(x) && \text{since } f'(u) = 5u^4 \\ &= 5(\sin(x))^4 \cdot \cos(x) \end{aligned}$$

Notice that it is quite easy to extend this to any power. Set $f(u) = u^n$. Then follow the same steps and we arrive at

$$F(x) = (\sin(x))^n \qquad F'(x) = n(\sin(x))^{n-1} \cos(x)$$

Example 2.9.5

This example shows one of the ways that the chain rule appears very frequently — when we need to differentiate the power of some simpler function. More generally we have the following.

Example 2.9.6

Let $f(u) = u^n$ and let $g(x)$ be any differentiable function. Set $F(x) = f(g(x)) = g(x)^n$. Then

$$\frac{dF}{dx} = \frac{d}{dx}(g(x)^n) = ng(x)^{n-1} \cdot g'(x)$$

This is precisely the result in Example 2.6.6 and Lemma 2.6.7.

Example 2.9.6

³³ In this context du is called a differential. There are ways to understand and manipulate these in calculus but they are beyond the scope of this course.

Example 2.9.7

Let $f(u) = \cos(u)$ and $g(x) = 3x - 2$. Find the derivative of

$$F(x) = f(g(x)) = \cos(3x - 2).$$

Again we should approach this by first writing down f and g and their derivatives and then putting everything together as the chain rule tells us.

$$\begin{aligned} f(u) &= \cos(u) \\ g(x) &= 3x - 2 \end{aligned}$$

$$\begin{aligned} f'(u) &= -\sin(u) \\ g'(x) &= 3 \end{aligned}$$

So the chain rule says

$$\begin{aligned} F'(x) &= f'(g(x)) \cdot g'(x) \\ &= -\sin(g(x)) \cdot 3 \\ &= -3\sin(3x - 2) \end{aligned}$$

Example 2.9.7

This example shows a second way that the chain rule appears very frequently — when we need to differentiate some function of $ax + b$. More generally we have the following.

Example 2.9.8

Let $a, b \in \mathbb{R}$ and let $f(x)$ be a differentiable function. Set $g(x) = ax + b$. Then

$$\begin{aligned} \frac{d}{dx}f(ax + b) &= \frac{d}{dx}f(g(x)) \\ &= f'(g(x)) \cdot g'(x) \\ &= f'(ax + b) \cdot a \end{aligned}$$

So the derivative of $f(ax + b)$ with respect to x is just $af'(ax + b)$.

Example 2.9.8

The above is a very useful result that follows from the chain rule, so let's make it a corollary to highlight it.

Corollary 2.9.9.

Let $a, b \in \mathbb{R}$ and let $f(x)$ be a differentiable function, then

$$\frac{d}{dx}f(ax + b) = af'(ax + b).$$

Example 2.9.10 (Example 2.9.1, continued)

Let us now go back to our motivating campfire example. There we had

$$\begin{aligned} f(x) &= \text{temperature at position } x \\ x(t) &= \text{position at time } t \\ F(t) &= f(x(t)) = \text{temperature at time } t \end{aligned}$$

The chain rule gave

$$F'(t) = f'(x(t)) \cdot x'(t)$$

Notice that the units of measurement on both sides of the equation agree — as indeed they must. To see this, let us assume that t is measured in seconds, that $x(t)$ is measured in metres and that $f(x)$ is measured in degrees. Because of this $F(x(t))$ must also be measured in degrees (since it is a temperature).

What about the derivatives? These are rates of change. So

- $F'(t)$ has units $\frac{\text{degrees}}{\text{second}}$,
- $f'(x)$ has units $\frac{\text{degrees}}{\text{metre}}$, and
- $x'(t)$ has units $\frac{\text{metre}}{\text{second}}$.

Hence the product

$$f'(x(t)) \cdot x'(t) \text{ has units } = \frac{\text{degrees}}{\text{metre}} \cdot \frac{\text{metre}}{\text{second}} = \frac{\text{degrees}}{\text{second}}.$$

has the same units as $F'(t)$. So the units on both sides of the equation agree. Checking that the units on both sides of an equation agree is a good check of consistency, but of course it does not prove that both sides are in fact the same.

Example 2.9.10

► (optional) — Derivation of the Chain Rule

First, let's review what our goal is. We have been given a function $g(x)$, that is differentiable at some point $x = a$, and another function $f(u)$, that is differentiable at the point $u = b = g(a)$. We have defined the composite function $F(x) = f(g(x))$ and we wish to show that

$$F'(a) = f'(g(a)) \cdot g'(a)$$

Before we can compute $F'(a)$, we need to set up some ground work, and in particular the definitions of our given derivatives:

$$f'(b) = \lim_{H \rightarrow 0} \frac{f(b+H) - f(b)}{H} \quad \text{and} \quad g'(a) = \lim_{h \rightarrow 0} \frac{g(a+h) - g(a)}{h}.$$

We are going to use similar manipulation tricks as we did back in the proofs of the arithmetic of derivatives in Section 2.5. Unfortunately, we have already used up the symbols “ F ” and “ H ”, so we are going to make use the Greek letters γ, φ .

As was the case in our derivation of the product rule it is convenient to introduce a couple of new functions. Set

$$\varphi(H) = \frac{f(b+H) - f(b)}{H}$$

Then we have

$$\lim_{H \rightarrow 0} \varphi(H) = f'(b) = f'(g(a)) \quad \text{since } b = g(a), \quad (2.9.1)$$

and we can also write (with a little juggling)

$$f(b+H) = f(b) + H\varphi(H)$$

Similarly set

$$\gamma(h) = \frac{g(a+h) - g(a)}{h}$$

which gives us

$$\lim_{h \rightarrow 0} \gamma(h) = g'(a) \quad \text{and} \quad g(a+h) = g(a) + h\gamma(h).$$

Now we can start computing

$$\begin{aligned} F'(a) &= \lim_{h \rightarrow 0} \frac{F(a+h) - F(a)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(g(a+h)) - f(g(a))}{h} \end{aligned}$$

We know that $g(a) = b$ and $g(a+h) = g(a) + h\gamma(h)$, so

$$\begin{aligned} F'(a) &= \lim_{h \rightarrow 0} \frac{f(g(a) + h\gamma(h)) - f(g(a))}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(b + h\gamma(h)) - f(b)}{h} \end{aligned}$$

Now for the sneaky bit. We can turn $f(b + h\gamma(h))$ into $f(b + H)$ by setting

$$H = h\gamma(h)$$

Now notice that as $h \rightarrow 0$ we have

$$\begin{aligned} \lim_{h \rightarrow 0} H &= \lim_{h \rightarrow 0} h \cdot \gamma(h) \\ &= \lim_{h \rightarrow 0} h \cdot \lim_{h \rightarrow 0} \gamma(h) \\ &= 0 \cdot g'(a) = 0 \end{aligned}$$

So as $h \rightarrow 0$ we also have $H \rightarrow 0$.

We now have

$$\begin{aligned}
 F'(a) &= \lim_{h \rightarrow 0} \frac{f(b+H) - f(b)}{h} \\
 &= \lim_{h \rightarrow 0} \underbrace{\frac{f(b+H) - f(b)}{H}}_{=\varphi(H)} \cdot \underbrace{\frac{H}{h}}_{=\gamma(h)} && \text{if } H = h\gamma(h) \neq 0 \\
 &= \lim_{h \rightarrow 0} (\varphi(H) \cdot \gamma(h)) \\
 &= \lim_{h \rightarrow 0} \varphi(H) \cdot \lim_{h \rightarrow 0} \gamma(h) && \text{since } H \rightarrow 0 \text{ as } h \rightarrow 0 \\
 &= \lim_{H \rightarrow 0} \varphi(H) \cdot \lim_{h \rightarrow 0} \gamma(h) && = f'(b) \cdot g'(a)
 \end{aligned}$$

This is exactly the RHS of the chain rule. It is possible to have $H = 0$ in the second line above. But that possibility is easy to deal with:

- If $g'(a) \neq 0$, then, since $\lim_{h \rightarrow 0} \gamma(h) = g'(a)$, $H = h\gamma(h)$ cannot be 0 for small nonzero h . Technically, there is an $h_0 > 0$ such that $H = h\gamma(h) \neq 0$ for all $0 < |h| < h_0$. In taking the limit $h \rightarrow 0$, above, we need only consider $0 < |h| < h_0$ and so, in this case, the above computation is completely correct.
- If $g'(a) = 0$, the above computation is still fine provided we exclude all h 's for which $H = h\gamma(h) \neq 0$. When $g'(a) = 0$, the right hand side, $f'(g(a)) \cdot g'(a)$, of the chain rule is 0. So the above computation gives

$$\lim_{\substack{h \rightarrow 0 \\ \gamma(h) \neq 0}} \frac{f(b+H) - f(b)}{h} = f'(g(a)) \cdot g'(a) = 0$$

On the other hand, when $H = 0$, we have $f(b+H) - f(b) = 0$. So

$$\lim_{\substack{h \rightarrow 0 \\ \gamma(h) = 0}} \frac{f(b+H) - f(b)}{h} = 0$$

too. That's all we need.

► Chain Rule Examples

We'll now use the chain rule to compute some more derivatives.

Example 2.9.11

Find $\frac{d}{dx}(1+3x)^{75}$.

This is a concrete version of Example [2.9.8](#). We are to find the derivative of a function that is built up by first computing $1+3x$ and then taking the 75th power of the result. So we set

$$\begin{aligned}
 f(u) &= u^{75} & f'(u) &= 75u^{74} \\
 g(x) &= 1+3x & g'(x) &= 3 \\
 F(x) &= f(g(x)) = g(x)^{75} = (1+3x)^{75}
 \end{aligned}$$

By the chain rule

$$\begin{aligned} F'(x) &= f'(g(x)) g'(x) = 75 g(x)^{74} g'(x) = 75 (1 + 3x)^{74} \cdot 3 \\ &= 225 (1 + 3x)^{74} \end{aligned}$$

Example 2.9.11

Example 2.9.12

Find $\frac{d}{dx} \sin(x^2)$.

In this example we are to compute the derivative of $\sin$ with a (slightly) complicated argument. So we apply the chain rule with f being $\sin$ and $g(x)$ being the complicated argument. That is, we set

$$\begin{aligned} f(u) &= \sin u & f'(u) &= \cos u \\ g(x) &= x^2 & g'(x) &= 2x \\ F(x) &= f(g(x)) = \sin(g(x)) = \sin(x^2) \end{aligned}$$

By the chain rule

$$\begin{aligned} F'(x) &= f'(g(x)) g'(x) = \cos(g(x)) g'(x) = \cos(x^2) \cdot 2x \\ &= 2x \cos(x^2) \end{aligned}$$

Example 2.9.12

Example 2.9.13

Find $\frac{d}{dx} \sqrt[3]{\sin(x^2)}$.

In this example we are to compute the derivative of the cube root of a (moderately) complicated argument, namely $\sin(x^2)$. So we apply the chain rule with f being “cube root” and $g(x)$ being the complicated argument. That is, we set

$$\begin{aligned} f(u) &= \sqrt[3]{u} = u^{\frac{1}{3}} & f'(u) &= \frac{1}{3} u^{-\frac{2}{3}} \\ g(x) &= \sin(x^2) & g'(x) &= 2x \cos(x^2) \\ F(x) &= f(g(x)) = \sqrt[3]{g(x)} = \sqrt[3]{\sin(x^2)} \end{aligned}$$

In computing $g'(x)$ here, we have already used the chain rule once (in Example 2.9.12). By the chain rule

$$\begin{aligned} F'(x) &= f'(g(x)) g'(x) = \frac{1}{3} g(x)^{-\frac{2}{3}} \cdot 2x \cos(x^2) \\ &= \frac{2x}{3} \frac{\cos(x^2)}{[\sin(x^2)]^{\frac{2}{3}}} \end{aligned}$$

Example 2.9.13

Example 2.9.14

Find the derivative of $\frac{d}{dx}f(g(h(x)))$.

This is very similar to the previous example. Let us set $F(x) = f(g(h(x)))$ with $u = g(h(x))$ then the chain rule tells us

$$\begin{aligned}\frac{dF}{dx} &= \frac{df}{du} \cdot \frac{du}{dx} \\ &= f'(g(h(x))) \cdot \frac{d}{dx}g(h(x))\end{aligned}$$

We now just apply the chain rule again

$$= f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x).$$

Indeed it is not too hard to generalise further (in the manner of Example 2.6.6 to find the derivative of the composition of 4 or more functions (though things start to become tedious to write down):

$$\begin{aligned}\frac{d}{dx}f_1(f_2(f_3(f_4(x)))) &= f'_1(f_2(f_3(f_4(x)))) \cdot \frac{d}{dx}f_2(f_3(f_4(x))) \\ &= f'_1(f_2(f_3(f_4(x)))) \cdot f'_2(f_3(f_4(x))) \cdot \frac{d}{dx}f_3(f_4(x)) \\ &= f'_1(f_2(f_3(f_4(x)))) \cdot f'_2(f_3(f_4(x))) \cdot f'_3(f_4(x)) \cdot f'_4(x)\end{aligned}$$

Example 2.9.14

Example 2.9.15

We can also use the chain rule to recover Corollary 2.4.6 and from there we can use the product rule to recover the quotient rule.

We want to differentiate $F(x) = \frac{1}{g(x)}$ so set $f(u) = \frac{1}{u}$ and $u = g(x)$. Then the chain rule tells us

$$\begin{aligned}\frac{d}{dx} \left\{ \frac{1}{g(x)} \right\} &= \frac{dF}{dx} = \frac{df}{du} \cdot \frac{du}{dx} \\ &= \frac{-1}{u^2} \cdot g'(x) \\ &= -\frac{g'(x)}{g(x)^2}.\end{aligned}$$

Once we know this, a quick application of the product rule will give us the quotient rule.

$$\begin{aligned}
 \frac{d}{dx} \left\{ \frac{f(x)}{g(x)} \right\} &= \frac{d}{dx} \left\{ f(x) \cdot \frac{1}{g(x)} \right\} && \text{use PR} \\
 &= f'(x) \cdot \frac{1}{g(x)} + f(x) \cdot \frac{d}{dx} \left\{ \frac{1}{g(x)} \right\} && \text{use the result from above} \\
 &= f'(x) \cdot \frac{1}{g(x)} - f(x) \cdot \frac{g'(x)}{g(x)^2} && \text{place over a common denominator} \\
 &= \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{g(x)^2}
 \end{aligned}$$

which is exactly the quotient rule.

Example 2.9.15

Example 2.9.16

Compute the following derivative:

$$\frac{d}{dx} \cos \left(\frac{x^5 \sqrt{3+x^6}}{(4+x^2)^3} \right)$$

This time we are to compute the derivative of $\cos$ with a really complicated argument.

- So, to start, we apply the chain rule with $g(x) = \frac{x^5 \sqrt{3+x^6}}{(4+x^2)^3}$ being the really complicated argument and f being $\cos$. That is, $f(u) = \cos(u)$. Since $f'(u) = -\sin(u)$, the chain rule gives

$$\frac{d}{dx} \cos \left(\frac{x^5 \sqrt{3+x^6}}{(4+x^2)^3} \right) = -\sin \left(\frac{x^5 \sqrt{3+x^6}}{(4+x^2)^3} \right) \frac{d}{dx} \left\{ \frac{x^5 \sqrt{3+x^6}}{(4+x^2)^3} \right\}$$

- This reduced our problem to that of computing the derivative of the really complicated argument $\frac{x^5 \sqrt{3+x^6}}{(4+x^2)^3}$. We can think of the argument as being built up out of three pieces, namely x^5 , multiplied by $\sqrt{3+x^6}$, divided by $(4+x^2)^3$, or, equivalently, multiplied by $(4+x^2)^{-3}$. So we may rewrite $\frac{x^5 \sqrt{3+x^6}}{(4+x^2)^3}$ as $x^5 (3+x^6)^{1/2} (4+x^2)^{-3}$, and then apply the product rule to reduce the problem to that of computing the derivatives of the three pieces.
- Here goes (recall Example 2.6.6):

$$\begin{aligned}
 \frac{d}{dx} [x^5 (3+x^6)^{1/2} (4+x^2)^{-3}] &= \frac{d}{dx} [x^5] \cdot (3+x^6)^{1/2} \cdot (4+x^2)^{-3} \\
 &\quad + x^5 \cdot \frac{d}{dx} [(3+x^6)^{1/2}] \cdot (4+x^2)^{-3} \\
 &\quad + x^5 \cdot (3+x^6)^{1/2} \cdot \frac{d}{dx} [(4+x^2)^{-3}]
 \end{aligned}$$

Q[13](*) : For what values of x does the derivative of $\frac{\sqrt{x}}{1-x^2}$ exist? Explain your answer.

Q[14]: Differentiate $f(x) = (3\sqrt[5]{x} + 15\sqrt[3]{x} + 8)(3x^2 + 8x - 5)$.

Q[15]: Differentiate $f(x) = \frac{(x^2 + 5x + 1)(\sqrt{x} + \sqrt[3]{x})}{x}$.

Q[16]: Find all x -values where $f(x) = \frac{1}{\frac{1}{5}x^5 + x^4 - \frac{5}{3}x^3}$ has a horizontal tangent line.

►► Stage 3

Q[17](*) : Find an equation of a line that is tangent to both of the curves $y = x^2$ and $y = x^2 - 2x + 2$ (at different points).

Q[18](*) : Find all lines that are tangent to both of the curves $y = x^2$ and $y = -x^2 + 2x - 5$. Illustrate your answer with a sketch.

Q[19](*) : Evaluate $\lim_{x \rightarrow 2} \left(\frac{x^{2015} - 2^{2015}}{x - 2} \right)$.

2.7▲ Derivatives of Exponential Functions

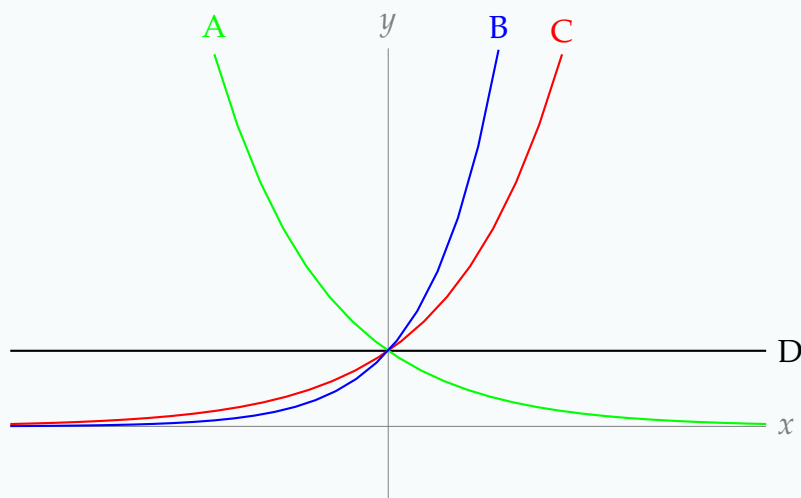
Exercises

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►► Stage 1

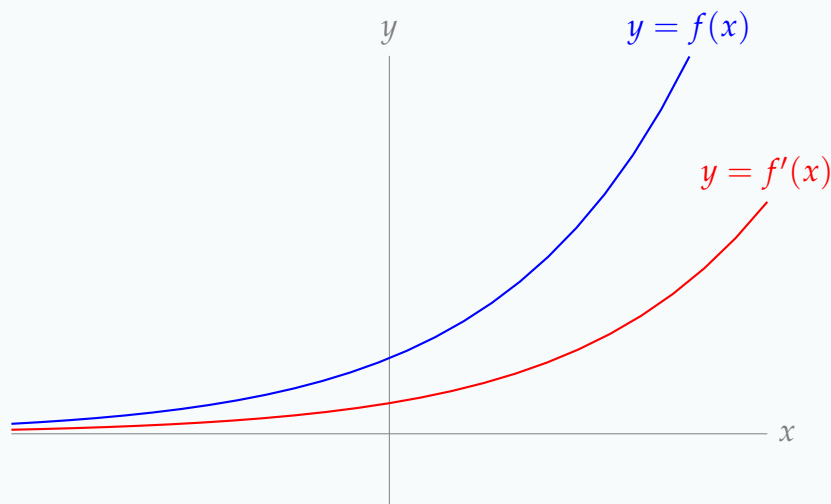
Q[1]: Match the curves in the graph to the following functions:

(a) $y = \left(\frac{1}{2}\right)^x$ (b) $y = 1^x$ (c) $y = 2^x$ (d) $y = 2^{-x}$ (e) $y = 3^x$



Q[2]: The graph below shows an exponential function $f(x) = a^x$ and its derivative $f'(x)$. Choose all the options that describe the constant a .

(a) $a < 0$ (b) $a > 0$ (c) $a < 1$ (d) $a > 1$ (e) $a < e$ (f) $a > e$



Q[3]: True or false: $\frac{d}{dx}\{e^x\} = xe^{x-1}$

Q[4]: A population of bacteria is described by $P(t) = 100e^{0.2t}$, for $0 \leq t \leq 10$. Over this time period, is the population increasing or decreasing?

We will learn more about the uses of exponential functions to describe real-world phenomena in Section 3.3.

►► Stage 2

Q[5]: Find the derivative of $f(x) = \frac{e^x}{2x}$.

Q[6]: Differentiate $f(x) = e^{2x}$.

Q[7]: Differentiate $f(x) = e^{a+x}$, where a is a constant.

Q[8]: For which values of x is the function $f(x) = xe^x$ increasing?

Q[9]: Differentiate e^{-x} .

Q[10]: Differentiate $f(x) = (e^x + 1)(e^x - 1)$.

Q[11]: A particle's position is given by

$$s(t) = t^2 e^t.$$

When is the particle moving in the negative direction?

►► Stage 3

Q[12]: Let $g(x) = f(x)e^x$, for a differentiable function $f(x)$. Give a simplified formula for $g'(x)$.

Functions of the form $g(x)$ are relatively common. If you remember this formula, you can save yourself some time when you need to differentiate them. We will explore this more in Question [19](#), Section 2.14.

Q[13]: Which of the following functions describe a straight line?

$$(a) y = e^{3 \log x} + 1 \quad (b) 2y + 5 = e^{3 + \log x} \quad (c) y = e^{2x} + 4 \quad (d) y = e^{\log x} 3^e + \log 2$$

Q[14](*) : Find constants a, b so that the following function is differentiable:

$$f(x) = \begin{cases} ax^2 + b & x \leq 1 \\ e^x & x > 1 \end{cases}$$

2.8▲ Derivatives of trigonometric functions

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

►► Stage 1

Q[1]: Graph sine and cosine on the same axes, from $x = -2\pi$ to $x = 2\pi$. Mark the points where $\sin x$ has a horizontal tangent. What do these points correspond to, on the graph of cosine?

Q[2]: Graph sine and cosine on the same axes, from $x = -2\pi$ to $x = 2\pi$. Mark the points where $\sin x$ has a tangent line of maximum (positive) slope. What do these points correspond to, on the graph of cosine?

►► Stage 2

Q[3]: Differentiate $f(x) = \sin x + \cos x + \tan x$.

Q[4]: For which values of x does the function $f(x) = \sin x + \cos x$ have a horizontal tangent?

Q[5]: Differentiate $f(x) = \sin^2 x + \cos^2 x$.

Q[6]: Differentiate $f(x) = 2 \sin x \cos x$.

Q[7]: Differentiate $f(x) = e^x \cot x$.

Q[8]: Differentiate $f(x) = \frac{2 \sin x + 3 \tan x}{\cos x + \tan x}$

Q[9]: Differentiate $f(x) = \frac{5 \sec x + 1}{e^x}$.

Q[10]: Differentiate $f(x) = (e^x + \cot x)(5x^6 - \csc x)$.

Q[11]: Differentiate $f(\theta) = \sin\left(\frac{\pi}{2} - \theta\right)$.

Q[12]: Differentiate $f(x) = \sin(-x) + \cos(-x)$.

Q[13]: Differentiate $s(\theta) = \frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta}$.

Q[14](*) : Find the values of the constants a and b for which

$$f(x) = \begin{cases} \cos(x) & x \leq 0 \\ ax + b & x > 0 \end{cases}$$

is differentiable everywhere.

Q[15](*) : Find the equation of the line tangent to the graph of $y = \cos(x) + 2x$ at $x = \frac{\pi}{2}$.

►► Stage 3

Q[16](*) : Evaluate $\lim_{x \rightarrow 2015} \left(\frac{\cos(x) - \cos(2015)}{x - 2015} \right)$.

Q[17](*) : Evaluate $\lim_{x \rightarrow \pi/3} \left(\frac{\cos(x) - 1/2}{x - \pi/3} \right)$.

Q[18](*) : Evaluate $\lim_{x \rightarrow \pi} \left(\frac{\sin(x)}{x - \pi} \right)$.

Q[19]: Show how you can use the quotient rule to find the derivative of tangent, if you already know the derivatives of sine and cosine.

Q[20](*) : The derivative of the function

$$f(x) = \begin{cases} ax + b & \text{for } x < 0 \\ \frac{6 \cos x}{2 + \sin x + \cos x} & \text{for } x \geq 0 \end{cases}$$

exists for all x . Determine the values of the constants a and b .

Q[21](*) : For which values of x does the derivative of $f(x) = \tan x$ exist?

Q[22](*) : For what values of x does the derivative of $\frac{10 \sin(x)}{x^2 + x - 6}$ exist? Explain your answer.

Q[23](*) : For what values of x does the derivative of $\frac{x^2 + 6x + 5}{\sin(x)}$ exist? Explain your answer.

Q[24](*) : Find the equation of the line tangent to the graph of $y = \tan(x)$ at $x = \frac{\pi}{4}$.

Q[25](*) : Find the equation of the line tangent to the graph of $y = \sin(x) + \cos(x) + e^x$ at $x = 0$.

Q[26]: For which values of x does the function $f(x) = e^x \sin x$ have a horizontal tangent line?

Q[27]: Let

$$f(x) = \begin{cases} \frac{\sin x}{x} & , \quad x \neq 0 \\ 1 & , \quad x = 0 \end{cases}$$

Find $f'(0)$, or show that it does not exist.

►► Application of Understanding

Q[28](*) : Differentiate the function

$$h(x) = \sin(|x|)$$

and give the domain where the derivative exists.

Q[29](*) : For the function

$$f(x) = \begin{cases} 0 & x \leq 0 \\ \frac{\sin(x)}{\sqrt{x}} & x > 0 \end{cases}$$

which of the following statements is correct?

- i. f is undefined at $x = 0$.
- ii. f is neither continuous nor differentiable at $x = 0$.
- iii. f is continuous but not differentiable at $x = 0$.
- iv. f is differentiable but not continuous at $x = 0$.
- v. f is both continuous and differentiable at $x = 0$.

Q[30](*) : Evaluate $\lim_{x \rightarrow 0} \frac{\sin x^{27} + 2x^5 e^{x^{99}}}{\sin^5 x}$.

2.9▲ One more tool - the chain rule

Exercises

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► Stage 1

Q[1]: Suppose the amount of kelp in a harbour depends on the number of urchins. Urchins eat kelp: when there are more urchins, there is less kelp, and when there are fewer urchins, there is more kelp. Suppose further that the number of urchins in the harbour depends on the number of otters, who find urchins extremely tasty: the more otters there are, the fewer urchins there are.

Let O , U , and K be the populations of otters, urchins, and kelp, respectively.

- (a) Is $\frac{dK}{dU}$ positive or negative?
- (b) Is $\frac{dU}{dO}$ positive or negative?
- (c) Is $\frac{dK}{dO}$ positive or negative?

Remark: An urchin barren is an area where unchecked sea urchin grazing has decimated the kelp population, which in turn causes the other species that shelter in the kelp forests to leave. Introducing otters to urchin barrens is one intervention to increase biodiversity. A short video with a more complex view of otters and urchins in Canadian waters is available on YouTube: <https://youtu.be/ASJ82wyHisE>

Q[2]: Suppose A , B , C , D , and E are functions describing an interrelated system, with the following signs: $\frac{dA}{dB} > 0$, $\frac{dB}{dC} > 0$, $\frac{dC}{dD} < 0$, and $\frac{dD}{dE} > 0$. Is $\frac{dA}{dE}$ positive or negative?

►► Stage 2

Q[3]: Evaluate the derivative of $f(x) = \cos(5x + 3)$.

Q[4]: Evaluate the derivative of $f(x) = (x^2 + 2)^5$.

Q[5]: Evaluate the derivative of $T(k) = (4k^4 + 2k^2 + 1)^{17}$.

Q[6]: Evaluate the derivative of $f(x) = \sqrt{\frac{x^2 + 1}{x^2 - 1}}$.

Q[7]: Evaluate the derivative of $f(x) = e^{\cos(x^2)}$.

Q[8](*) : Evaluate $f'(2)$ if $f(x) = g(x/h(x))$, $h(2) = 2$, $h'(2) = 3$, $g'(1) = 4$.

Q[9](*) : Find the derivative of $e^{x \cos(x)}$.

Q[10](*) : Evaluate $f'(x)$ if $f(x) = e^{x^2 + \cos x}$.

Q[11](*) : Evaluate $f'(x)$ if $f(x) = \sqrt{\frac{x-1}{x+2}}$.

Q[12](*) : Differentiate the function

$$f(x) = \frac{1}{x^2} + \sqrt{x^2 - 1}$$

and give the domain where the derivative exists.

Q[13](*) : Evaluate the derivative of $f(x) = \frac{\sin 5x}{1 + x^2}$

Q[14]: Evaluate the derivative of $f(x) = \sec(e^{2x+7})$.

Q[15]: Find the tangent line to the curve $y = (\tan^2 x + 1)(\cos^2 x)$ at the point $x = \frac{\pi}{4}$.

Q[16]: The position of a particle at time t is given by $s(t) = e^{t^3 - 7t^2 + 8t}$. For which values of t is the velocity of the particle zero?

Q[17]: What is the slope of the tangent line to the curve $y = \tan(e^{x^2})$ at the point $x = 1$?

Q[18](*) : Differentiate $y = e^{4x} \tan x$. You do not need to simplify your answer.

Q[19](*) : Evaluate the derivative of the following function at $x = 1$: $f(x) = \frac{x^3}{1 + e^{3x}}$.

Q[20](*) : Differentiate $e^{\sin^2(x)}$.

Q[21](*) : Compute the derivative of $y = \sin(e^{5x})$

Q[22](*) : Find the derivative of $e^{\cos(x^2)}$.

Q[23](*) : Compute the derivative of $y = \cos(x^2 + \sqrt{x^2 + 1})$

Q[24](*) : Evaluate the derivative.

$$y = (1 + x^2) \cos^2 x$$

Q[25](*) : Evaluate the derivative.

$$y = \frac{e^{3x}}{1 + x^2}$$

Q[26](*) : Find $g'(2)$ if $g(x) = x^3 h(x^2)$, where $h(4) = 2$ and $h'(4) = -2$.

Q[27](*) : At what points (x, y) does the curve $y = xe^{-(x^2-1)/2}$ have a horizontal tangent?

Q[28]: A particle starts moving at time $t = 1$, and its position thereafter is given by

$$s(t) = \sin\left(\frac{1}{t}\right).$$

When is the particle moving in the negative direction?

Q[29]: Compute the derivative of $f(x) = \frac{e^x}{\cos^3(5x-7)}$.

Q[30](*) : Evaluate $\frac{d}{dx} \{xe^{2x} \cos 4x\}$.

►► Stage 3

Q[31]: A particle moves along the Cartesian plane from time $t = -\pi/2$ to time $t = \pi/2$. The x -coordinate of the particle at time t is given by $x = \cos t$, and the y -coordinate is given by $y = \sin t$, so the particle traces a curve in the plane. When does the tangent line to that curve have slope -1 ?

Q[32](*) : Show that, for all $x > 0$, $e^{x+x^2} > 1 + x$.

Q[33]: We know that $\sin(2x) = 2 \sin x \cos x$. What other trig identity can you derive from this, using differentiation?

S-19: This limit represents the derivative computed at $x = 2$ of the function $f(x) = x^{2015}$. To see this, simply use the definition of the derivative at $a = 2$ with $f(x) = x^{2015}$:

$$\left. \frac{d}{dx} \{f(x)\} \right|_a = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$\left. \frac{d}{dx} \{x^{2015}\} \right|_2 = \lim_{x \rightarrow 2} \frac{x^{2015} - 2^{2015}}{x - 2}$$

Since the derivative of $f(x)$ is $2015 \cdot x^{2014}$, then its value at $x = 2$ is exactly $2015 \cdot 2^{2014}$.

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S-1: Since $1^x = 1$ for any x , we see that (b) is just the constant function $y = 1$, so D matches to (b).

Since $2^{-x} = \frac{1}{2^x} = \left(\frac{1}{2}\right)^x$, functions (a) and (d) are the same. This is the only function out of the lot that grows as $x \rightarrow -\infty$ and shrinks as $x \rightarrow \infty$, so A matches to (a) and (d).

This leaves B and C to match to (c) and (e). Since $3 > 2$, when $x > 0$, $3^x > 2^x$. So, (e) matches to the function that grows more quickly to the right of the x -axis: B matches to (e), and C matches to (c).

S-2: First, let's consider the behaviour of exponential functions a^x based on whether a is greater or less than 1. As we know, $\lim_{x \rightarrow \infty} a^x = \begin{cases} \infty & a > 1 \\ 0 & a < 1 \end{cases}$ and $\lim_{x \rightarrow -\infty} a^x = \begin{cases} 0 & a > 1 \\ \infty & a < 1 \end{cases}$. Our function has $\lim_{x \rightarrow \infty} f(x) = \infty$ and $\lim_{x \rightarrow -\infty} f(x) = 0$, so we conclude $a > 1$: thus (d) and also (b) hold. (We could have also seen that (b) holds because a^x is defined for all real numbers.)

It remains to decide whether a is greater or less than e . (If a were equal to e , then $f'(x)$ would be the same as $f(x)$.) We saw in the text that $\frac{d}{dx} \{a^x\} = C(a)a^x$ for the function $C(a) = \lim_{h \rightarrow 0} \frac{a^h - 1}{h}$. We know that $C(e) = 1$. (Actually, we chose e to be the number that has this property.) From our graph, we see that $f'(x) < f(x)$, so $C(a) < 1 = C(e)$. In other words, $\lim_{h \rightarrow 0} \frac{a^h - 1}{h} < \lim_{h \rightarrow 0} \frac{e^h - 1}{h}$; so, $a < e$. Thus (e) holds.

S-3: The power rule tells us that $\frac{d}{dx} \{x^n\} = nx^{n-1}$. In this equation, the variable is the base, and the exponent is a constant. In the function e^x , it's reversed: the variable is the exponent, and the base is a constant. So, the power rule does not apply.

S-4: $P(t)$ is an increasing function over its domain, so the population is increasing.

There are a few ways to see that $P(t)$ is increasing.

What we really care about is whether $e^{0.2t}$ is increasing or decreasing, since an increasing function multiplied by 100 is still an increasing function, and a decreasing function multiplied by 100 is still a decreasing function. Since $f(t) = e^t$ is an increasing function, we can use what we know about graphing functions to see that $f(0.2t) = e^{0.2t}$ is also increasing.

S-5: Using the quotient rule,

$$f'(x) = \frac{2xe^x - 2e^x}{4x^2} = \frac{e^x(2x - 2)}{4x^2} = \frac{(x - 1)e^x}{2x^2}$$

S-6:

$$f'(x) = \frac{d}{dx}\{e^{2x}\} = \frac{d}{dx}\{(e^x)^2\} = 2\frac{d}{dx}\{e^x\}e^x = 2e^xe^x = 2(e^x)^2 = 2e^{2x}$$

S-7:

$$e^{a+x} = e^ae^x$$

Since e^a is just a constant,

$$\frac{d}{dx}\{e^ae^x\} = e^a\frac{d}{dx}\{e^x\} = e^ae^x = e^{a+x}$$

So, $f'(x) = f(x) = e^{a+x}$.

S-8: If the derivative is positive, the function is increasing, so let's start by finding the derivative. We use the product rule (although Question **12** gives a shortcut).

$$f'(x) = 1 \cdot e^x + xe^x = (1 + x)e^x$$

Since e^x is always positive, $f'(x) > 0$ when $1 + x > 0$. So, $f(x)$ is increasing when $x > -1$.

S-9:

$$e^{-x} = \frac{1}{e^x}$$

Using the rule for differentiating the reciprocal:

$$\frac{d}{dx}\{e^{-x}\} = \frac{-e^x}{(e^x)^2} = \frac{-1}{e^x} = -e^{-x}$$

S-10: Using the product rule,

$$f'(x) = (e^x)(e^x - 1) + (e^x + 1)(e^x) = e^x(e^x - 1 + e^x + 1) = 2(e^x)^2 = 2e^{2x}$$

Alternate solution: using Question **6**:

$$f(x) = e^{2x} - 1 \implies f'(x) = 2e^{2x}.$$

S-11: The question asks when $s'(t)$ is negative. So, we start by differentiating. Using the product rule:

$$\begin{aligned} s'(t) &= e^t(t^2 + 2t) \\ &= e^t \cdot t(t + 2) \end{aligned}$$

e^t is always positive, so $s'(t)$ is negative when t and $2 + t$ have opposite signs. This occurs when $-2 < t < 0$.

S-12: Using the product rule, $g'(x) = f'(x)e^x + f(x)e^x = [f(x) + f'(x)]e^x$

S-13: We simplify the functions to get a better idea of what's going on.

(a): $y = e^{3 \log x} + 1 = (e^{\log x})^3 + 1 = x^3 + 1$. This is not a line.

(b): $2y + 5 = e^{3 + \log x} = e^3 e^{\log x} = e^3 x$. Since e^3 is a constant, $2y + 5 = e^3 x$ is a line.

(c): There isn't a fancy simplification here—this isn't a line. If that isn't a satisfactory answer, we can check: a line is a function with a constant slope. For our function, $y' = \frac{d}{dx}\{e^{2x} + 4\} = \frac{d}{dx}\{e^{2x}\} = \frac{d}{dx}\{(e^x)^2\} = 2e^x e^x = 2e^{2x}$. Since the derivative isn't constant, the function isn't a line.

(d): $y = e^{\log x} 3^e + \log 2 = 3^e x + \log 2$. Since 3^e and $\log 2$ are constants, this is a line.

S-14: When we say a function is differentiable without specifying a range, we mean that it is differentiable over its domain. The function $f(x)$ is differentiable when $x \neq 1$ for any values of a and b ; it is up to us to figure out which constants make it differentiable when $x = 1$.

In order to be differentiable, a function must be continuous. The definition of continuity tells us that, for f to be continuous at $x = 1$, we need $\lim_{x \rightarrow 1} f(x) = f(1)$. From the definition of f , we see $f(1) = a + b = \lim_{x \rightarrow 1^-} f(x)$, so we need $\lim_{x \rightarrow 1^+} f(x) = a + b$. Since $\lim_{x \rightarrow 1^+} f(x) = e^1 = e$, we specifically need

$$e = a + b.$$

Now, let's consider differentiability of f at $x = 1$. We need the following limit to exist:

$$\lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$$

In particular, we need the one-sided limits to exist and be equal:

$$\lim_{h \rightarrow 0^-} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h}$$

If $h < 0$, then $1+h < 1$, so $f(1+h) = a(1+h)^2 + b$. If $h > 0$, then $1+h > 1$, so $f(1+h) = e^{1+h}$. With this in mind, we begin to evaluate the one-sided limits:

$$\begin{aligned} \lim_{h \rightarrow 0^-} \frac{f(1+h) - f(1)}{h} &= \lim_{h \rightarrow 0^-} \frac{[a(1+h)^2 + b] - [a + b]}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{ah^2 + 2ah}{h} = 2a \\ \lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h} &= \lim_{h \rightarrow 0^+} \frac{e^{1+h} - (a + b)}{h} \end{aligned}$$

Since we take $a + b$ to be equal to e (to ensure continuity):

$$\begin{aligned} &= \lim_{h \rightarrow 0^+} \frac{e^{1+h} - e^1}{h} \\ &= \left. \frac{d}{dx} \{e^x\} \right|_{x=1} = e^1 = e \end{aligned}$$

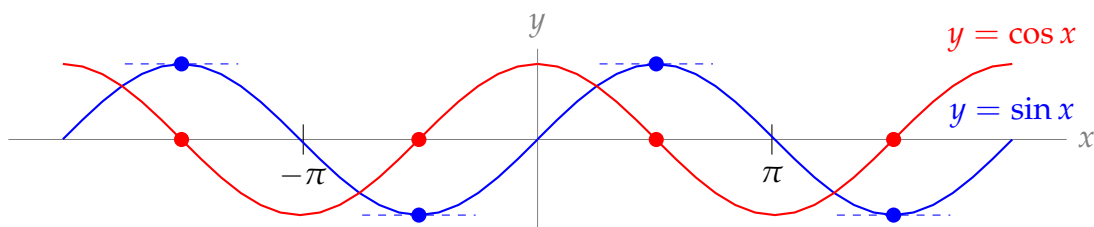
So, we also need

$$2a = e$$

Therefore, the values of a and b that make f differentiable are $a = b = \frac{e}{2}$.

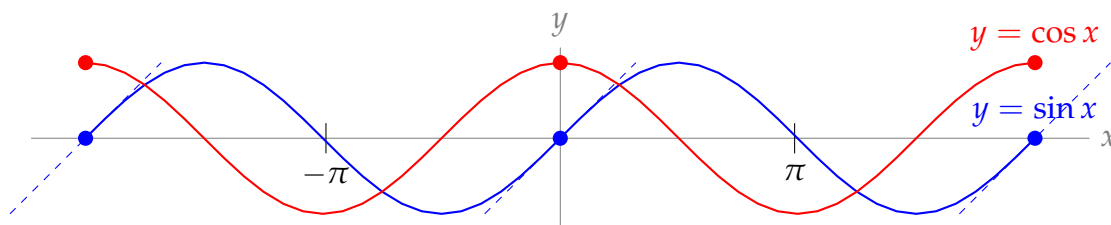
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S-1:



The graph $f(x) = \sin x$ has horizontal tangent lines precisely at those points where $\cos x = 0$. This must be true, since $\frac{d}{dx} \{\sin x\} = \cos x$: where the derivative of sine is zero, cosine itself is zero.

S-2:

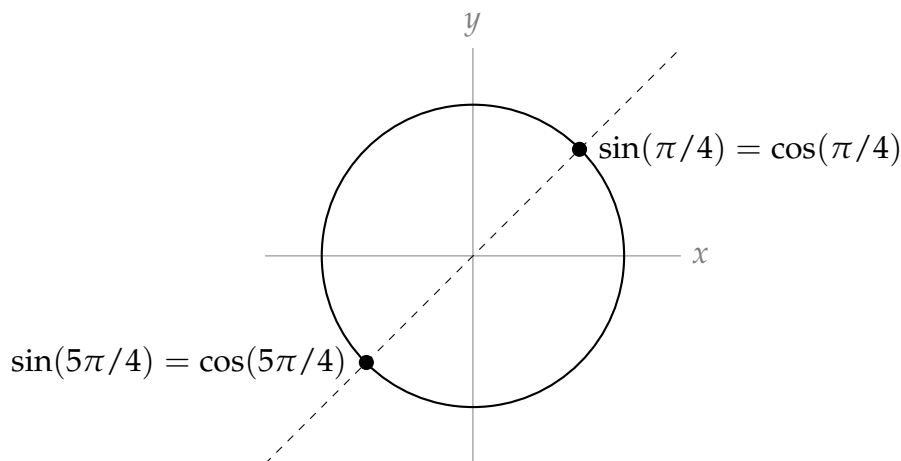


The graph $f(x) = \sin x$ has maximum slope at those points where $\cos x$ has a maximum. This makes sense, because $f'(x) = \cos x$: the maximum values of the slope of sine correspond to the maximum values of cosine.

S-3: You should memorize the derivatives of sine, cosine, and tangent.

$$f'(x) = \cos x - \sin x + \sec^2 x$$

S-4: $f'(x) = \cos x - \sin x$, so $f'(x) = 0$ precisely when $\sin x = \cos x$. This happens at $\pi/4$, but it also happens at $5\pi/4$. By looking at the unit circle, it is clear that $\sin x = \cos x$ whenever $x = \frac{\pi}{4} + \pi n$ for some integer n .



S-5:

- Solution 1: $f(x) = \sin^2 x + \cos^2 x = 1$, so $f'(x) = \frac{d}{dx}\{1\} = 0$.
- Solution 2: Using the formula for the derivative of a squared function,

$$f'(x) = 2 \sin x \cos x + 2 \cos x (-\sin x) = 2 \sin x \cos x - 2 \sin x \cos x = 0.$$

S-6: It is true that $2 \sin x \cos x = \sin(2x)$, but we don't know the derivative of $\sin(2x)$. So, we use the product rule:

$$f'(x) = 2 \cos x \cos x + 2 \sin x (-\sin x) = 2(\cos^2 x - \sin^2 x).$$

S-7:

- Solution 1: using the product rule,

$$f'(x) = e^x \cot x + e^x(-\csc^2 x) = e^x(\cot x - \csc^2 x).$$

- Solution 2: using the formula from Question **12**, Section 2.7,

$$f'(x) = e^x(\cot x - \csc^2 x).$$

S-8: We use the quotient rule.

$$\begin{aligned} f'(x) &= \frac{(\cos x + \tan x)(2 \cos x + 3 \sec^2 x) - (2 \sin x + 3 \tan x)(-\sin x + \sec^2 x)}{(\cos x + \tan x)^2} \\ &= \frac{2 \cos^2 x + 3 \cos x \sec^2 x + 2 \cos x \tan x + 3 \tan x \sec^2 x}{(\cos x + \tan x)^2} \\ &\quad + \frac{2 \sin^2 x - 2 \sin x \sec^2 x + 3 \sin x \tan x - 3 \tan x \sec^2 x}{(\cos x + \tan x)^2} \\ &= \frac{2 + 3 \sec x + 2 \sin x - 2 \tan x \sec x + 3 \sin x \tan x}{(\cos x + \tan x)^2} \end{aligned}$$

S-9: We use the quotient rule.

$$\begin{aligned} f'(x) &= \frac{e^x(5 \sec x \tan x) - (5 \sec x + 1)e^x}{(e^x)^2} \\ &= \frac{5 \sec x \tan x - 5 \sec x - 1}{e^x} \end{aligned}$$

S-10: We use the product rule:

$$f'(x) = (e^x + \cot x)(30x^5 + \csc x \cot x) + (e^x - \csc^2 x)(5x^6 - \csc x)$$

S-11: We don't know how to differentiate this function as it is written, but an identity helps us. Since $\sin(\frac{\pi}{2} - \theta) = \cos \theta$, we see $f'(\theta) = \frac{d}{d\theta}\{\cos \theta\} = -\sin(\theta)$.

S-12: We know the derivative of $\sin x$, but not of $\sin(-x)$. So we re-write $f(x)$ using identities:

$$\begin{aligned} f(x) &= \sin(-x) + \cos(-x) \\ &= -\sin x + \cos x \\ f'(x) &= -\cos x - \sin x \end{aligned}$$

S-13: We apply the quotient rule.

$$\begin{aligned}s'(\theta) &= \frac{(\cos \theta - \sin \theta)(-\sin \theta + \cos \theta) - (\cos \theta + \sin \theta)(-\sin \theta - \cos \theta)}{(\cos \theta - \sin \theta)^2} \\&= \frac{(\cos \theta - \sin \theta)^2 + (\cos \theta + \sin \theta)^2}{(\cos \theta - \sin \theta)^2} \\&= 1 + \left(\frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta} \right)^2\end{aligned}$$

S-14: In order for f to be differentiable at $x = 0$, it must also be continuous at $x = 0$. This forces

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0) \quad \text{or} \quad \lim_{x \rightarrow 0^-} \cos(x) = \lim_{x \rightarrow 0^+} (ax + b) = 1$$

or $b = 1$. In order for f to be differentiable at $x = 0$, we need the limit

$$\lim_{h \rightarrow 0} \frac{f(0 + h) - f(0)}{h}$$

to exist. This is the case if and only if the two one-sided limits

$$\lim_{h \rightarrow 0^-} \frac{f(0 + h) - f(0)}{h} = \lim_{h \rightarrow 0^-} \frac{\cos(h) - \cos(0)}{h}$$

and

$$\lim_{h \rightarrow 0^+} \frac{f(0 + h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{(ah + b) - \cos(0)}{h} = a \quad \text{since } b = 1$$

exist and are equal. Because $\cos(x)$ is differentiable at $x = 0$ we have

$$\lim_{h \rightarrow 0^-} \frac{\cos(h) - \cos(0)}{h} = \left. \frac{d}{dx} \cos(x) \right|_{x=0} = -\sin(x) \Big|_{x=0} = 0$$

So, we need $a = 0$ and $b = 1$.

S-15: We compute the derivative of $\cos(x) + 2x$ as being $-\sin(x) + 2$, which evaluated at $x = \frac{\pi}{2}$ yields $-1 + 2 = 1$. Since we also compute $\cos(\pi/2) + 2(\pi/2) = 0 + \pi$, then the equation of the tangent line is

$$y - \pi = 1 \cdot (x - \pi/2).$$

S-16: This limit represents the derivative computed at $x = 2015$ of the function $f(x) = \cos(x)$. To see this, simply use the definition of the derivative at $a = 2015$ with $f(x) = \cos x$:

$$\begin{aligned}\left. \frac{d}{dx} \{f(x)\} \right|_a &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \\ \left. \frac{d}{dx} \{\cos x\} \right|_{2015} &= \lim_{x \rightarrow 2015} \frac{\cos(x) - \cos(2015)}{x - 2015}\end{aligned}$$

Since the derivative of $f(x)$ is $-\sin(x)$, its value at $x = 2015$ is exactly $-\sin(2015)$.

S-17: This limit represents the derivative computed at $x = \pi/3$ of the function $f(x) = \cos x$. To see this, simply use the definition of the derivative at $a = \pi/3$ with $f(x) = \cos x$:

$$\begin{aligned}\left. \frac{d}{dx} \{f(x)\} \right|_a &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \\ \left. \frac{d}{dx} \{\cos x\} \right|_{\pi/3} &= \lim_{x \rightarrow \pi/3} \frac{\cos(x) - \cos(\pi/3)}{x - \pi/3} \\ &= \lim_{x \rightarrow \pi/3} \frac{\cos(x) - 1/2}{x - \pi/3}\end{aligned}$$

Since the derivative of $f(x)$ is $-\sin x$, then its value at $x = \pi/3$ is exactly $-\sin(\pi/3) = -\sqrt{3}/2$.

S-18: This limit represents the derivative computed at $x = \pi$ of the function $f(x) = \sin(x)$. To see this, simply use the definition of the derivative at $a = \pi$ with $f(x) = \sin x$:

$$\begin{aligned}\left. \frac{d}{dx} \{f(x)\} \right|_a &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \\ \left. \frac{d}{dx} \{\sin x\} \right|_{\pi} &= \lim_{x \rightarrow \pi} \frac{\sin(x) - \sin(\pi)}{x - \pi} \\ &= \lim_{x \rightarrow \pi} \frac{\sin(x)}{x - \pi}\end{aligned}$$

Since the derivative of $f(x)$ is $\cos(x)$, then its value at $x = \pi$ is exactly $\cos(\pi) = -1$.

S-19:

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

So, using the quotient rule,

$$\begin{aligned}\frac{d}{d\theta} \{\tan \theta\} &= \frac{\cos \theta \cos \theta - \sin \theta(-\sin \theta)}{\cos^2 \theta} = \frac{\cos^2 \theta + \sin^2 \theta}{\cos^2 \theta} \\ &= \left(\frac{1}{\cos \theta} \right)^2 = \sec^2 \theta\end{aligned}$$

S-20: In order for the function $f(x)$ to be continuous at $x = 0$, the left half formula $ax + b$ and the right half formula $\frac{6 \cos x}{2 + \sin x + \cos x}$ must match up at $x = 0$. This forces

$$a \times 0 + b = \frac{6 \cos 0}{2 + \sin 0 + \cos 0} = \frac{6}{3} \implies \boxed{b = 2}$$

In order for the derivative $f'(x)$ to exist at $x = 0$, the limit $\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$ must exist. In particular, the limits $\lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h}$ and $\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h}$ must exist and be equal to each other.

When $h \rightarrow 0^-$, this means $h < 0$, so $f(h) = ah + b = ah + 2$. So:

$$\lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0^-} \frac{(ah + 2) - 2}{h} = \frac{d}{dx} \{ax + 2\} \Big|_{x=0} = a.$$

Similarly, when $h \rightarrow 0^+$, then $h > 0$, so $f(h) = \frac{6 \cos h}{1 + \sin h + \cos h}$ and

$$\begin{aligned} \lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} &= \frac{d}{dx} \left\{ \frac{6 \cos x}{2 + \sin x + \cos x} \right\} \Big|_{x=0} \\ &= \frac{-6 \sin x(2 + \sin x + \cos x) - 6 \cos x(\cos x - \sin x)}{(2 + \sin x + \cos x)^2} \Big|_{x=0}. \end{aligned}$$

Since the limits from the left and right must be equal, this forces

$$a = \frac{-6 \sin 0(2 + \sin 0 + \cos 0) - 6 \cos 0(\cos 0 - \sin 0)}{(2 + \sin 0 + \cos 0)^2} = \frac{-6}{(2 + 1)^2} \implies \boxed{a = -\frac{2}{3}}$$

S-21: In order for $f'(x)$ to exist, $f(x)$ has to exist. We already know that $\tan x$ does not exist whenever $x = \frac{\pi}{2} + n\pi$ for any integer n . If we look a little deeper, since $\tan x = \frac{\sin x}{\cos x}$, the points where tangent does not exist correspond exactly to the points where cosine is zero.

From its graph, tangent looks like a smooth curve over its domain, so we might guess that everywhere tangent is defined, its derivative is defined. We can check this:

$f'(x) = \sec^2 x = \left(\frac{1}{\cos x} \right)^2$. Indeed, wherever $\cos x$ is nonzero, f' exists.

So, $f'(x)$ exists for all values of x *except* when $x = \frac{\pi}{2} + n\pi$ for some integer n .

S-22: The function is differentiable whenever $x^2 + x - 6 \neq 0$ since the derivative equals

$$\frac{10 \cos(x) \cdot (x^2 + x - 6) - 10 \sin(x) \cdot (2x + 1)}{(x^2 + x - 6)^2},$$

which is well-defined unless $x^2 + x - 6 = 0$. We solve $x^2 + x - 6 = (x - 2)(x + 3) = 0$, and get $x = 2$ and $x = -3$. So, the function is differentiable for all real values x except for $x = 2$ and for $x = -3$.

S-23: The function is differentiable whenever $\sin(x) \neq 0$ since the derivative equals

$$\frac{\sin(x) \cdot (2x + 6) - \cos(x) \cdot (x^2 + 6x + 5)}{(\sin x)^2},$$

which is well-defined unless $\sin x = 0$. This happens when x is an integer multiple of π . So, the function is differentiable for all real values x except $x = n\pi$, where n is any integer.

S-24: We compute the derivative of $\tan(x)$ as being $\sec^2(x)$, which evaluated at $x = \frac{\pi}{4}$ yields 2. Since we also compute $\tan(\pi/4) = 1$, then the equation of the tangent line is

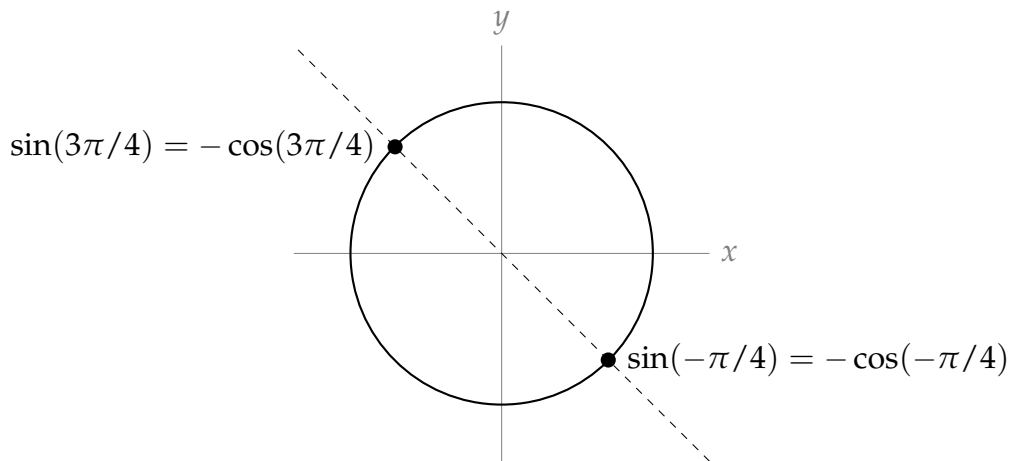
$$y - 1 = 2 \cdot (x - \pi/4).$$

S-25: We compute the derivative $y' = \cos(x) - \sin(x) + e^x$, which evaluated at $x = 0$ yields $1 - 0 + 1 = 2$. Since we also compute $y(0) = 0 + 1 + 1 = 2$, the equation of the tangent line is

$$y - 2 = 2(x - 0)$$

ie $y = 2x + 2$.

S-26: We are asked to solve $f'(x) = 0$. That is, $e^x[\sin x + \cos x] = 0$. Since e^x is always positive, that means we need to find all points where $\sin x + \cos x = 0$. That is, we need to find all values of x where $\sin x = -\cos x$. Looking at the unit circle, we see this happens whenever $x = \frac{3\pi}{4} + n\pi$ for any integer n .



S-27: First, we note that our function is continuous, because

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 = f(0)$$

This is a handy thing to check: if the function were discontinuous at $x = 0$, then we would automatically know that it was not differentiable there.

Now, on to the derivative. We can use the limit definition:

$$\begin{aligned} f'(0) &= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} && \text{if it exists} \\ &= \lim_{h \rightarrow 0} \frac{f(h) - 1}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{\sin h}{h} - 1}{h} \end{aligned}$$

As h approaches 0, both the numerator and the denominator approach 0. So, to evaluate the limit, we need to do more work. The key insight we can use is a result that was shown in the text while evaluating the derivative of sine. When h is close to 0, $\cos h \leq \frac{\sin h}{h} \leq 1$. We use this to bound our limit, and then apply the squeeze theorem.

$$\begin{aligned} \cos h &\leq \frac{\sin h}{h} \leq 1 \\ \text{So, } \cos h - 1 &\leq \frac{\sin h}{h} - 1 \leq 1 - 1 = 0 \end{aligned}$$

and $\frac{1}{h} \left(\frac{\sin h}{h} - 1 \right)$ is between 0 and $\frac{\cos h - 1}{h}$. We can evaluate the limit of $\frac{\cos h - 1}{h}$, by noticing its similarity to the definition of the derivative.

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} &= \lim_{h \rightarrow 0} \frac{\cos(0+h) - \cos(0)}{h} \\ &= \frac{d}{dx} \{\cos x\}|_{x=0} = 0 \end{aligned}$$

So, by the Squeeze Theorem, $f'(0) = \lim_{h \rightarrow 0} \frac{\frac{\sin h}{h} - 1}{h} = 0$.

S-28: As usual, when dealing with the absolute value function, we can make things a little clearer by splitting it up into two pieces.

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

So,

$$\sin |x| = \begin{cases} \sin x & x \geq 0 \\ \sin(-x) & x < 0 \end{cases} = \begin{cases} \sin x & x \geq 0 \\ -\sin x & x < 0 \end{cases}$$

where we used the identity $\sin(-x) = -\sin x$. From here, it's easy to see $h'(x)$ when x is anything *other than* zero.

$$\frac{d}{dx}\{\sin |x|\} = \begin{cases} \cos x & x > 0 \\ ?? & x = 0 \\ -\cos x & x < 0 \end{cases}$$

To decide whether $h(x)$ is differentiable at $x = 0$, we use the definition of the derivative. One word of explanation: usually in the definition of the derivative, h is the tiny “change in x ” that is going to zero. Since h is the name of our function, we need another letter to stand for the tiny change in x , the size of which is tending to zero. We chose t .

$$\lim_{t \rightarrow 0} \frac{h(t+0) - h(0)}{t} = \lim_{t \rightarrow 0} \frac{\sin |t|}{t}$$

We consider the behaviour of this function to the left and right of $t = 0$:

$$\frac{\sin |t|}{t} = \begin{cases} \frac{\sin t}{t} & t \geq 0 \\ \frac{\sin(-t)}{t} & t < 0 \end{cases} = \begin{cases} \frac{\sin t}{t} & t \geq 0 \\ -\frac{\sin t}{t} & t < 0 \end{cases}$$

Since we're evaluating the limit as t goes to zero, we need the fact that $\lim_{t \rightarrow 0} \frac{\sin t}{t} = 1$. We saw this in Section 2.7, but also we know enough now to evaluate it another way. Using the definition of the derivative:

$$\lim_{t \rightarrow 0} \frac{\sin t}{t} = \lim_{t \rightarrow 0} \frac{\sin(t+0) - \sin(0)}{t} = \left. \frac{d}{dx}\{\sin x\} \right|_{t=0} = \cos 0 = 1$$

At any rate, since we know $\lim_{t \rightarrow 0} \frac{\sin t}{t} = 1$, then:

$$\lim_{t \rightarrow 0^+} \frac{h(t+0) - h(0)}{t} = \lim_{t \rightarrow 0^+} \frac{\sin t}{t} = 1 \quad \lim_{t \rightarrow 0^-} \frac{h(t+0) - h(0)}{t} = \lim_{t \rightarrow 0^-} \frac{-\sin t}{t} = -1$$

So, since the one-sided limits disagree,

$$\lim_{t \rightarrow 0} \frac{h(t+0) - h(0)}{t} = DNE$$

so $h(x)$ is not differentiable at $x = 0$. Therefore,

$$h'(x) = \begin{cases} \cos x & x > 0 \\ -\cos x & x < 0 \end{cases}$$

S-29: Statement **i** is false, since $f(0) = 0$. Statement **iv** cannot hold, since a function that is differentiable is also continuous.

Since $\lim_{x \rightarrow 0^+} \frac{\sin x}{x} = 1$ (we saw this in Section 2.8),

$$\begin{aligned} \lim_{x \rightarrow 0^+} f(x) &= \lim_{x \rightarrow 0^+} \frac{\sin x}{\sqrt{x}} \\ &= \lim_{x \rightarrow 0^+} \sqrt{x} \frac{\sin x}{x} \\ &= 0 \cdot 1 = 0 \end{aligned}$$

So f is continuous at $x = 0$, and so Statement [ii](#) does not hold. Now, let's consider $f'(x)$.

$$\begin{aligned}\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x} &= \lim_{x \rightarrow 0^+} \frac{\frac{\sin x}{\sqrt{x}} - 0}{x} \\ &= \lim_{x \rightarrow 0^+} \frac{1}{\sqrt{x}} \frac{\sin x}{x} = +\infty\end{aligned}$$

Therefore, using the definition of the derivative,

$$\begin{aligned}f'(0) &= \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} \quad \text{if it exists, but} \\ \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} &= DNE\end{aligned}$$

since one of the one-sided limits does not exist. So f is continuous but not differentiable at $x = 0$. The correct statement is [iii](#).

S-30: Recall that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$. In order to take advantage of this knowledge, we divide the numerator and denominator by x^5 (because 5 is the power of sine in the denominator, and a denominator that goes to zero generally makes a limit harder).

$$\lim_{x \rightarrow 0} \frac{\sin x^{27} + 2x^5 e^{x^{99}}}{\sin^5 x} = \lim_{x \rightarrow 0} \frac{\frac{\sin x^{27}}{x^5} + 2e^{x^{99}}}{\left(\frac{\sin x}{x}\right)^5}$$

Now the denominator goes to 1, which is nice, but we need to take care of the fraction $\frac{\sin x^{27}}{x^5}$ in the numerator. This fraction isn't very familiar, but we know that, as x goes to zero, x^{27} also goes to zero, so that $\frac{\sin x^{27}}{x^{27}}$ goes to 1. Consequently,

$$\lim_{x \rightarrow 0} \frac{\sin x^{27} + 2x^5 e^{x^{99}}}{\sin^5 x} = \lim_{x \rightarrow 0} \frac{x^{22} \frac{\sin x^{27}}{x^{27}} + 2e^{x^{99}}}{\left(\frac{\sin x}{x}\right)^5} = \frac{0 \times 1 + 2 \times e^0}{1^5} = 2$$

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S-1: [\(a\)](#) More urchins means less kelp, and fewer urchins means more kelp. This means kelp and urchins are negatively correlated, so $\frac{dK}{dU} < 0$.

If you aren't sure why that is, we give a more detailed explanation here, using the definition of the derivative. When h is a positive number, $U + h$ is greater than U , so $K(U + h)$ is less than $K(U)$, hence $K(U + h) - K(U) < 0$. Therefore:

$$\lim_{h \rightarrow 0^+} \frac{K(U + h) - K(U)}{h} = \frac{\text{negative}}{\text{positive}} < 0.$$

Similarly, when h is negative, $U + h$ is less than U , so $K(U + h) - K(U) > 0$, and

$$\lim_{h \rightarrow 0^-} \frac{K(U + h) - K(U)}{h} = \frac{\text{positive}}{\text{negative}} < 0.$$

Therefore:

$$\frac{dK}{dU} = \lim_{h \rightarrow 0} \frac{K(U + h) - K(U)}{h} < 0.$$

(b) More otters means fewer urchins, and fewer otters means more urchins. So, otters and urchins are negatively correlated: $\frac{dU}{dO} < 0$.

(c) Using the chain rule, $\frac{dK}{dO} = \frac{dK}{dU} \cdot \frac{dU}{dO}$. Parts (a) and (b) tell us both these derivatives are negative, so their product is positive: $\frac{dK}{dO} > 0$.

We can also see that $\frac{dK}{dO} > 0$ by thinking about the relationships as described. When the otter population increases, the urchin population decreases, so the kelp population increases. That means when the otter population increases, the kelp population also increases, so kelp and otters are positively correlated. The chain rule is a formal version of this kind of reasoning.

S-2:

$$\frac{dA}{dE} = \frac{dA}{dB} \cdot \frac{dB}{dC} \cdot \frac{dC}{dD} \cdot \frac{dD}{dE} < 0$$

since we multiply three positive quantities and one negative.

S-3: Applying the chain rule:

$$\begin{aligned} \frac{d}{dx} \{\cos(5x + 3)\} &= -\sin(5x + 3) \cdot \frac{d}{dx} \{5x + 3\} \\ &= -\sin(5x + 3) \cdot 5 \end{aligned}$$

S-4: Using the chain rule,

$$\begin{aligned} f'(x) &= \frac{d}{dx} \left\{ (x^2 + 2)^5 \right\} \\ &= 5 (x^2 + 2)^4 \cdot \frac{d}{dx} \{x^2 + 2\} \\ &= 5(x^2 + 2)^4 \cdot 2x \\ &= 10x(x^2 + 2)^4 \end{aligned}$$

S-5: Using the chain rule,

$$\begin{aligned}T'(k) &= \frac{d}{dk} \left\{ (4k^4 + 2k^2 + 1)^{17} \right\} \\&= 17(4k^4 + 2k^2 + 1)^{16} \cdot \frac{d}{dk} \{4k^4 + 2k^2 + 1\} \\&= 17(4k^4 + 2k^2 + 1)^{16} \cdot (16k^3 + 4k)\end{aligned}$$

S-6: Using the chain rule:

$$\begin{aligned}\frac{d}{dx} \left\{ \sqrt{\frac{x^2 + 1}{x^2 - 1}} \right\} &= \frac{1}{2\sqrt{\frac{x^2 + 1}{x^2 - 1}}} \cdot \frac{d}{dx} \left\{ \frac{x^2 + 1}{x^2 - 1} \right\} \\&= \frac{1}{2} \sqrt{\frac{x^2 - 1}{x^2 + 1}} \cdot \frac{d}{dx} \left\{ \frac{x^2 + 1}{x^2 - 1} \right\}\end{aligned}$$

And now, the quotient rule:

$$\begin{aligned}&= \frac{1}{2} \sqrt{\frac{x^2 - 1}{x^2 + 1}} \cdot \left(\frac{(x^2 - 1)(2x) - (x^2 + 1)2x}{(x^2 - 1)^2} \right) \\&= \frac{1}{2} \sqrt{\frac{x^2 - 1}{x^2 + 1}} \cdot \left(\frac{-4x}{(x^2 - 1)^2} \right) \\&= \sqrt{\frac{x^2 - 1}{x^2 + 1}} \cdot \left(\frac{-2x}{(x^2 - 1)^2} \right) \\&= \frac{-2x}{(x^2 - 1)\sqrt{x^4 - 1}}\end{aligned}$$

S-7: If we let $g(x) = e^x$ and $h(x) = \cos(x^2)$, then $f(x) = g(h(x))$, so $f'(x) = g'(h(x)) \cdot h'(x)$.

$$f'(x) = e^{\cos(x^2)} \cdot \frac{d}{dx} \{\cos(x^2)\}$$

In order to evaluate $\frac{d}{dx} \{\cos(x^2)\}$, we'll need the chain rule *again*.

$$\begin{aligned}&= e^{\cos(x^2)} \cdot [-\sin(x^2)] \cdot \frac{d}{dx} \{x^2\} \\&= -e^{\cos(x^2)} \cdot \sin(x^2) \cdot 2x\end{aligned}$$

S-8: We use the chain rule, followed by the quotient rule:

$$\begin{aligned} f'(x) &= g' \left(\frac{x}{h(x)} \right) \cdot \frac{d}{dx} \left\{ \frac{x}{h(x)} \right\} \\ &= g' \left(\frac{x}{h(x)} \right) \cdot \frac{h(x) - xh'(x)}{h(x)^2} \end{aligned}$$

When $x = 2$:

$$\begin{aligned} f'(2) &= g' \left(\frac{2}{h(2)} \right) \frac{h(2) - 2h'(2)}{h(2)^2} \\ &= 4 \frac{2 - 2 \times 3}{2^2} = -4 \end{aligned}$$

S-9: Using the chain rule, followed by the product rule:

$$\begin{aligned} \frac{d}{dx} \left\{ e^{x \cos(x)} \right\} &= e^{x \cos x} \frac{d}{dx} \{ x \cos x \} \\ &= [\cos x - x \sin x] e^{x \cos(x)} \end{aligned}$$

S-10: Using the chain rule:

$$\begin{aligned} \frac{d}{dx} \left\{ e^{x^2 + \cos(x)} \right\} &= e^{x^2 + \cos x} \frac{d}{dx} \{ x^2 + \cos x \} \\ &= [2x - \sin x] e^{x^2 + \cos(x)} \end{aligned}$$

S-11: Using the chain rule, followed by the quotient rule:

$$\begin{aligned} \frac{d}{dx} \left\{ \sqrt{\frac{x-1}{x+2}} \right\} &= \frac{1}{2\sqrt{\frac{x-1}{x+2}}} \frac{d}{dx} \left\{ \frac{x-1}{x+2} \right\} \\ &= \frac{\sqrt{x+2}}{2\sqrt{x-1}} \cdot \frac{(x+2) - (x-1)}{(x+2)^2} \\ &= \frac{3}{2\sqrt{x-1}\sqrt{x+2}^3} \end{aligned}$$

S-12: First, we manipulate our function to make it easier to differentiate:

$$f(x) = x^{-2} + (x^2 - 1)^{1/2}$$

Now, we can use the power rule to differentiate $\frac{1}{x^2}$. This will be easier than differentiating $\frac{1}{x^2}$ using quotient rule, but if you prefer, quotient rule will also work.

$$\begin{aligned} f'(x) &= -2x^{-3} + \frac{1}{2}(x^2 - 1)^{-1/2} \cdot \frac{d}{dx}\{x^2 - 1\} \\ &= -2x^{-3} + \frac{1}{2}(x^2 - 1)^{-1/2}(2x) \\ &= \frac{-2}{x^3} + \frac{x}{\sqrt{x^2 - 1}} \end{aligned}$$

The function $f(x)$ is only defined when $x \neq 0$ and when $x^2 - 1 \geq 0$. That is, when x is in $(-\infty, -1] \cup [1, \infty)$. We have an added restriction on the domain of $f'(x)$: $x^2 - 1$ must not be zero. So, the domain of $f'(x)$ is $(-\infty, -1) \cup (1, \infty)$.

S-13: We use the quotient rule, noting that $\frac{d}{dx}\{\sin 5x\} = 5 \cos 5x$:

$$f'(x) = \frac{(1 + x^2)(5 \cos 5x) - (\sin 5x)(2x)}{(1 + x^2)^2}$$

S-14: If we let $g(x) = \sec x$ and $h(x) = e^{2x+7}$, then $f(x) = g(h(x))$, so by the chain rule, $f'(x) = g'(h(x)) \cdot h'(x)$. Since $g'(x) = \sec x \tan x$:

$$\begin{aligned} f'(x) &= g'(h(x)) \cdot h'(x) \\ &= \sec(h(x)) \tan(h(x)) \cdot h'(x) \\ &= \sec(e^{2x+7}) \tan(e^{2x+7}) \cdot \frac{d}{dx}\{e^{2x+7}\} \end{aligned}$$

Here, we need the chain rule again:

$$\begin{aligned} &= \sec(e^{2x+7}) \tan(e^{2x+7}) \cdot \left[e^{2x+7} \cdot \frac{d}{dx}\{2x + 7\} \right] \\ &= \sec(e^{2x+7}) \tan(e^{2x+7}) \cdot \left[e^{2x+7} \cdot 2 \right] \\ &= 2e^{2x+7} \sec(e^{2x+7}) \tan(e^{2x+7}) \end{aligned}$$

S-15: It is possible to start in on this problem with the product rule and then the chain rule, but it's easier if we simplify first. Since $\tan^2 x + 1 = \sec^2 x = \frac{1}{\cos^2 x}$, we see

$$f(x) = \frac{\cos^2 x}{\cos^2 x} = 1$$

for all values of x for which $\cos x$ is nonzero. That is, $f(x) = 1$ for every x that is not an integer multiple of $\pi/2$ (and $f(x)$ is not defined when x is an integer multiple of $\pi/2$).

Therefore, $f'(x) = 0$ for every x on which f exists, and in particular $f'(\pi/4) = 0$. Also, $f(\pi/4) = 1$, so the tangent line to f at $x = \pi/4$ is the line with slope 0, passing through the point $(\pi/4, 1)$:

$$y = 1$$

S-16: Velocity is the derivative of position with respect to time. So, the velocity of the particle is given by $s'(t)$. We need to find $s'(t)$, and determine when it is zero.

To differentiate, we use the chain rule.

$$\begin{aligned} s'(t) &= e^{t^3-7t^2+8t} \cdot \frac{d}{dt}\{t^3 - 7t^2 + 8t\} \\ &= e^{t^3-7t^2+8t} \cdot (3t^2 - 14t + 8) \end{aligned}$$

To determine where this function is zero, we factor:

$$= e^{t^3-7t^2+8t} \cdot (3t - 2)(t - 4)$$

So, the velocity is zero when $e^{t^3-7t^2+8t} = 0$, when $3t - 2 = 0$, and when $t - 4 = 0$. Since $e^{t^3-7t^2+8t}$ is *never* zero, this tells us that the velocity is zero precisely when $t = \frac{2}{3}$ or $t = 4$.

S-17: The slope of the tangent line is the derivative. If we let $f(x) = \tan x$ and $g(x) = e^{x^2}$, then $f(g(x)) = \tan(e^{x^2})$, so $y' = f'(g(x)) \cdot g'(x)$:

$$y' = \sec^2(e^{x^2}) \cdot \frac{d}{dx}\{e^{x^2}\}$$

We find ourselves once more in need of the chain rule:

$$\begin{aligned} &= \sec^2(e^{x^2}) \cdot e^{x^2} \frac{d}{dx}\{x^2\} \\ &= \sec^2(e^{x^2}) \cdot e^{x^2} \cdot 2x \end{aligned}$$

Finally, we evaluate this derivative at the point $x = 1$:

$$\begin{aligned} y'(1) &= \sec^2(e) \cdot e \cdot 2 \\ &= 2e \sec^2 e \end{aligned}$$

S-18: Using the Product rule,

$$y' = \frac{d}{dx}\{e^{4x}\} \tan x + e^{4x} \sec^2 x$$

and the chain rule:

$$\begin{aligned} &= e^{4x} \cdot \frac{d}{dx}\{4x\} \cdot \tan x + e^{4x} \sec^2 x \\ &= 4e^{4x} \tan x + e^{4x} \sec^2 x \end{aligned}$$

S-19: Using the quotient rule,

$$f'(x) = \frac{(3x^2)(1 + e^{3x}) - (x^3) \cdot \frac{d}{dx}\{1 + e^{3x}\}}{(1 + e^{3x})^2}$$

Now, the chain rule:

$$= \frac{(3x^2)(1 + e^{3x}) - (x^3)(3e^{3x})}{(1 + e^{3x})^2}$$

So, when $x = 1$:

$$f'(1) = \frac{3(1 + e^3) - 3e^3}{(1 + e^3)^2} = \frac{3}{(1 + e^3)^2}$$

S-20: This requires us to apply the chain rule twice.

$$\begin{aligned}\frac{d}{dx} \{e^{\sin^2(x)}\} &= e^{\sin^2(x)} \cdot \frac{d}{dx} \{\sin^2(x)\} \\ &= e^{\sin^2(x)} (2 \sin(x)) \cdot \frac{d}{dx} \sin(x) \\ &= e^{\sin^2(x)} (2 \sin(x)) \cdot \cos(x)\end{aligned}$$

S-21: This requires us to apply the chain rule twice.

$$\begin{aligned}\frac{d}{dx} \{\sin(e^{5x})\} &= \cos(e^{5x}) \cdot \frac{d}{dx} \{e^{5x}\} \\ &= \cos(e^{5x}) (e^{5x}) \cdot \frac{d}{dx} \{5x\} \\ &= \cos(e^{5x}) (e^{5x}) \cdot 5\end{aligned}$$

S-22: We'll use the chain rule twice.

$$\begin{aligned}\frac{d}{dx} \{e^{\cos(x^2)}\} &= e^{\cos(x^2)} \cdot \frac{d}{dx} \{\cos(x^2)\} \\ &= e^{\cos(x^2)} \cdot (-\sin(x^2)) \cdot \frac{d}{dx} \{x^2\} \\ &= -e^{\cos(x^2)} \cdot \sin(x^2) \cdot 2x\end{aligned}$$

S-23: We start with the chain rule:

$$\begin{aligned}y' &= -\sin(x^2 + \sqrt{x^2 + 1}) \cdot \frac{d}{dx} \{x^2 + \sqrt{x^2 + 1}\} \\&= -\sin(x^2 + \sqrt{x^2 + 1}) \cdot \left(2x + \frac{d}{dx} \{\sqrt{x^2 + 1}\}\right)\end{aligned}$$

and find ourselves in need of chain rule a second time:

$$\begin{aligned}&= -\sin(x^2 + \sqrt{x^2 + 1}) \cdot \left(2x + \frac{1}{2\sqrt{x^2 + 1}} \cdot \frac{d}{dx} \{x^2 + 1\}\right) \\&= -\sin(x^2 + \sqrt{x^2 + 1}) \cdot \left(2x + \frac{2x}{2\sqrt{x^2 + 1}}\right)\end{aligned}$$

S-24:

$$y = (1 + x^2) \cos^2 x$$

Using the product rule,

$$y' = (2x) \cos^2 x + (1 + x^2) \frac{d}{dx} \{\cos^2 x\}$$

Here, we'll need to use the chain rule. Remember $\cos^2 x = [\cos x]^2$.

$$\begin{aligned}&= 2x \cos^2 x + (1 + x^2) 2 \cos x \cdot \frac{d}{dx} \{\cos x\} \\&= 2x \cos^2 x + (1 + x^2) 2 \cos x \cdot (-\sin x) \\&= 2x \cos^2 x - 2(1 + x^2) \sin x \cos x\end{aligned}$$

S-25: We use the quotient rule, noting by the chain rule that $\frac{d}{dx} \{e^{3x}\} = 3e^{3x}$:

$$\begin{aligned}y' &= \frac{(1 + x^2) \cdot 3e^{3x} - e^{3x}(2x)}{(1 + x^2)^2} \\&= \frac{e^{3x}(3x^2 - 2x + 3)}{(1 + x^2)^2}\end{aligned}$$

S-26: By the chain rule,

$$\frac{d}{dx} \{h(x^2)\} = h'(x^2) \cdot 2x$$

Using the product rules and the result above,

$$g'(x) = 3x^2 h(x^2) + x^3 h'(x^2) 2x$$

Plugging in $x = 2$:

$$\begin{aligned} g'(2) &= 3(2^2)h(2^2) + 2^3h'(2^2)2 \times 2 \\ &= 12h(4) + 32h'(4) = 12 \times 2 - 32 \times 2 \\ &= -40 \end{aligned}$$

S-27: Let $f(x) = xe^{-(x^2-1)/2} = xe^{(1-x^2)/2}$. Then, using the product rule,

$$f'(x) = e^{(1-x^2)/2} + x \cdot \frac{d}{dx} \left\{ e^{(1-x^2)/2} \right\}$$

Here, we need the chain rule:

$$\begin{aligned} &= e^{(1-x^2)/2} + x \cdot e^{(1-x^2)/2} \frac{d}{dx} \left\{ \frac{1}{2}(1-x^2) \right\} \\ &= e^{(1-x^2)/2} + x \cdot e^{(1-x^2)/2} \cdot (-x) \\ &= (1-x^2)e^{(1-x^2)/2} \end{aligned}$$

There is no power of e that is equal to zero; so if the product above is zero, it must be that $1-x^2 = 0$. This happens for $x = \pm 1$. On the curve, when $x = 1$, $y = 1$, and when $x = -1$, $y = -1$. So the points are $(1, 1)$ and $(-1, -1)$.

S-28: The question asks when $s'(t)$ is negative. So, we start by differentiating. Using the chain rule:

$$\begin{aligned} s'(t) &= \cos\left(\frac{1}{t}\right) \cdot \frac{d}{dt} \left\{ \frac{1}{t} \right\} \\ &= \cos\left(\frac{1}{t}\right) \cdot \frac{-1}{t^2} \end{aligned}$$

When $t \geq 1$, $\frac{1}{t}$ is between 0 and 1. Since $\cos \theta$ is positive for $0 \leq \theta < \pi/2$, and $\pi/2 > 1$, we see that $\cos\left(\frac{1}{t}\right)$ is positive for the entire domain of $s(t)$. Also, $\frac{-1}{t^2}$ is negative for the entire domain of the function. We conclude that $s'(t)$ is negative for the entire domain of $s(t)$, so the particle is *always* moving in the negative direction.

S-29: We present two solutions: one where we dive right in and use the quotient rule, and another where we simplify first and use the product rule.

- Solution 1: We begin with the quotient rule:

$$\begin{aligned} f'(x) &= \frac{\cos^3(5x-7) \frac{d}{dx}\{e^x\} - e^x \frac{d}{dx}\{\cos^3(5x-7)\}}{\cos^6(5x-7)} \\ &= \frac{\cos^3(5x-7)e^x - e^x \frac{d}{dx}\{\cos^3(5x-7)\}}{\cos^6(5x-7)} \end{aligned}$$

Now, we use the chain rule. Since $\cos^3(5x - 7) = [\cos(5x - 7)]^3$, our “outside” function is $g(x) = x^3$, and our “inside” function is $h(x) = \cos(5x - 7)$.

$$= \frac{\cos^3(5x - 7)e^x - e^x \cdot 3\cos^2(5x - 7) \cdot \frac{d}{dx}\{\cos(5x - 7)\}}{\cos^6(5x - 7)}$$

We need the chain rule again!

$$\begin{aligned} &= \frac{\cos^3(5x - 7)e^x - e^x \cdot 3\cos^2(5x - 7) \cdot [-\sin(5x - 7) \cdot \frac{d}{dx}\{5x - 7\}]}{\cos^6(5x - 7)} \\ &= \frac{\cos^3(5x - 7)e^x - e^x \cdot 3\cos^2(5x - 7) \cdot [-\sin(5x - 7) \cdot 5]}{\cos^6(5x - 7)} \end{aligned}$$

We finish by simplifying:

$$\begin{aligned} &= \frac{e^x \cos^2(5x - 7) (\cos(5x - 7) + 15 \sin(5x - 7))}{\cos^6(5x - 7)} \\ &= e^x \frac{\cos(5x - 7) + 15 \sin(5x - 7)}{\cos^4(5x - 7)} \\ &= e^x (\sec^3(5x - 7) + 15 \tan(5x - 7) \sec^3(5x - 7)) \\ &= e^x \sec^3(5x - 7) (1 + 15 \tan(5x - 7)) \end{aligned}$$

- Solution 2: We simplify to avoid the quotient rule:

$$\begin{aligned} f(x) &= \frac{e^x}{\cos^3(5x - 7)} \\ &= e^x \sec^3(5x - 7) \end{aligned}$$

Now we use the product rule to differentiate:

$$f'(x) = e^x \sec^3(5x - 7) + e^x \frac{d}{dx}\{\sec^3(5x - 7)\}$$

Here, we'll need the chain rule. Since $\sec^3(5x - 7) = [\sec(5x - 7)]^3$, our “outside” function is $g(x) = x^3$ and our “inside” function is $h(x) = \sec(5x - 7)$, so that $g(h(x)) = [\sec(5x - 7)]^3 = \sec^3(5x - 7)$.

$$= e^x \sec^3(5x - 7) + e^x \cdot 3 \sec^2(5x - 7) \cdot \frac{d}{dx}\{\sec(5x - 7)\}$$

We need the chain rule again! Recall $\frac{d}{dx}\{\sec x\} = \sec x \tan x$.

$$\begin{aligned} &= e^x \sec^3(5x - 7) + e^x \cdot 3 \sec^2(5x - 7) \cdot \sec(5x - 7) \tan(5x - 7) \cdot \frac{d}{dx}\{5x - 7\} \\ &= e^x \sec^3(5x - 7) + e^x \cdot 3 \sec^2(5x - 7) \cdot \sec(5x - 7) \tan(5x - 7) \cdot 5 \end{aligned}$$

We finish by simplifying:

$$= e^x \sec^3(5x - 7) (1 + 15 \tan(5x - 7))$$

S-30:

- Solution 1: In Example 2.6.6, we generalized the product rule to three factors:

$$\frac{d}{dx}\{f(x)g(x)h(x)\} = f'(x)g(x)h(x) + f(x)g'(x)h(x) + f(x)g(x)h'(x)$$

Using this rule:

$$\begin{aligned}\frac{d}{dx}\{(x)(e^{2x})(\cos 4x)\} &= \frac{d}{dx}\{x\} \cdot e^{2x} \cos 4x + x \cdot \frac{d}{dx}\{e^{2x}\} \cdot \cos 4x + xe^{2x} \cdot \frac{d}{dx}\{\cos 4x\} \\ &= e^{2x} \cos 4x + x(2e^{2x}) \cos 4x + xe^{2x}(-4 \sin 4x) \\ &= e^{2x} \cos 4x + 2xe^{2x} \cos 4x - 4xe^{2x} \sin 4x\end{aligned}$$

- Solution 2: We can use the product rule twice. In the first step, we split the function $xe^{2x} \cos 4x$ into the product of two functions.

$$\begin{aligned}\frac{d}{dx}\{(xe^{2x})(\cos 4x)\} &= \frac{d}{dx}\{xe^{2x}\} \cdot \cos 4x + xe^{2x} \cdot \frac{d}{dx}\{\cos 4x\} \\ &= \left(\frac{d}{dx}\{x\} \cdot e^{2x} + x \cdot \frac{d}{dx}\{e^{2x}\}\right) \cdot \cos 4x + xe^{2x} \cdot \frac{d}{dx}\{\cos 4x\} \\ &= (e^{2x} + x(2e^{2x})) \cdot \cos 4x + xe^{2x}(-4 \sin 4x) \\ &= e^{2x} \cos 4x + 2xe^{2x} \cos 4x - 4xe^{2x} \sin 4x\end{aligned}$$

S-31: At time t , the particle is at the point $(x(t), y(t))$, with $x(t) = \cos t$ and $y(t) = \sin t$. Over time, the particle traces out a curve; let's call that curve $y = f(x)$. Then $y(t) = f(x(t))$, so the slope of the curve at the point $(x(t), y(t))$ is $f'(x(t))$. You are to determine the values of t for which $f'(x(t)) = -1$.

By the chain rule

$$y'(t) = f'(x(t)) \cdot x'(t)$$

Substituting in $x(t) = \cos t$ and $y(t) = \sin t$ gives

$$\cos t = f'(x(t)) \cdot (-\sin t)$$

so that

$$f'(x(t)) = -\frac{\cos t}{\sin t}$$

is -1 precisely when $\sin t = \cos t$. This happens whenever $t = \frac{\pi}{4}$.

Remark: the path traced by the particle is a semicircle. You can think about the point on the unit circle with angle t , or you can notice that $x^2 + y^2 = \sin^2 t + \cos^2 t = 1$.

S-32: Let $f(x) = e^{x+x^2}$ and $g(x) = 1 + x$. Then $f(0) = g(0) = 1$.

$f'(x) = (1 + 2x)e^{x+x^2}$ and $g'(x) = 1$. When $x > 0$,

$$f'(x) = (1 + 2x)e^{x+x^2} > 1 \cdot e^{x+x^2} = e^{x+x^2} > e^{0+0^2} = 1 = g'(x).$$

Since $f(0) = g(0)$, and $f'(x) > g'(x)$ for all $x > 0$, that means f and g start at the same place, but f always grows faster. Therefore, $f(x) > g(x)$ for all $x > 0$.

S-33: Since $\sin 2x$ and $2 \sin x \cos x$ are the same function, they have the same derivative.

$$\begin{aligned}\sin 2x &= 2 \sin x \cos x \\ \Rightarrow \frac{d}{dx} \{\sin 2x\} &= \frac{d}{dx} \{2 \sin x \cos x\} \\ 2 \cos 2x &= 2[\cos^2 x - \sin^2 x] \\ \cos 2x &= \cos^2 x - \sin^2 x\end{aligned}$$

We conclude $\cos 2x = \cos^2 x - \sin^2 x$, which is another common trig identity.

Remark: if we differentiate both sides of this equation, we get the original identity back.

S-34:

$$\begin{aligned}f(x) &= \sqrt[3]{\frac{e^{\csc x^2}}{\sqrt{x^3 - 9 \tan x}}} \\ &= \left(\frac{e^{\csc x^2}}{\sqrt{x^3 - 9 \tan x}} \right)^{\frac{1}{3}}\end{aligned}$$

To begin the differentiation, we can choose our “outside” function to be $g(x) = x^{\frac{1}{3}}$, and our “inside” function to be $h(x) = \frac{e^{\csc x^2}}{\sqrt{x^3 - 9 \tan x}}$. Then $f(x) = g(h(x))$, so $f'(x) = g'(h(x)) \cdot h'(x) = \frac{1}{3}(h(x))^{-\frac{2}{3}}h'(x)$:

$$\begin{aligned}f'(x) &= \frac{1}{3} \left(\frac{e^{\csc x^2}}{\sqrt{x^3 - 9 \tan x}} \right)^{-\frac{2}{3}} \cdot \frac{d}{dx} \left\{ \frac{e^{\csc x^2}}{\sqrt{x^3 - 9 \tan x}} \right\} \\ &= \frac{1}{3} \left(\frac{\sqrt{x^3 - 9 \tan x}}{e^{\csc x^2}} \right)^{\frac{2}{3}} \cdot \frac{d}{dx} \left\{ \frac{e^{\csc x^2}}{\sqrt{x^3 - 9 \tan x}} \right\}\end{aligned}$$

This leads us to use the quotient rule:

$$= \frac{1}{3} \left(\frac{\sqrt{x^3 - 9 \tan x}}{e^{\csc x^2}} \right)^{\frac{2}{3}} \left(\frac{\sqrt{x^3 - 9 \tan x} \frac{d}{dx} \{e^{\csc x^2}\} - e^{\csc x^2} \frac{d}{dx} \{\sqrt{x^3 - 9 \tan x}\}}{(\tan^2 x)(x^3 - 9)} \right)$$

Let's figure out those two derivatives on their own, then plug them in. Using the chain rule twice:

$$\begin{aligned}\frac{d}{dx} \left\{ e^{\csc x^2} \right\} &= e^{\csc x^2} \frac{d}{dx} \left\{ \csc x^2 \right\} = e^{\csc x^2} \cdot (-\csc(x^2) \cot(x^2)) \cdot \frac{d}{dx} \{x^2\} \\ &= -2xe^{\csc x^2} \csc(x^2) \cot(x^2)\end{aligned}$$

For the other derivative, we start with the product rule, then chain:

$$\begin{aligned}\frac{d}{dx} \left\{ \sqrt{x^3 - 9} \tan x \right\} &= \frac{d}{dx} \left\{ \sqrt{x^3 - 9} \right\} \cdot \tan x + \sqrt{x^3 - 9} \sec^2 x \\ &= \frac{1}{2\sqrt{x^3 - 9}} \frac{d}{dx} \left\{ x^3 - 9 \right\} \cdot \tan x + \sqrt{x^3 - 9} \sec^2 x \\ &= \frac{3x^2 \tan x}{2\sqrt{x^3 - 9}} + \sqrt{x^3 - 9} \sec^2 x\end{aligned}$$

Now, we plug these into our equation for $f'(x)$:

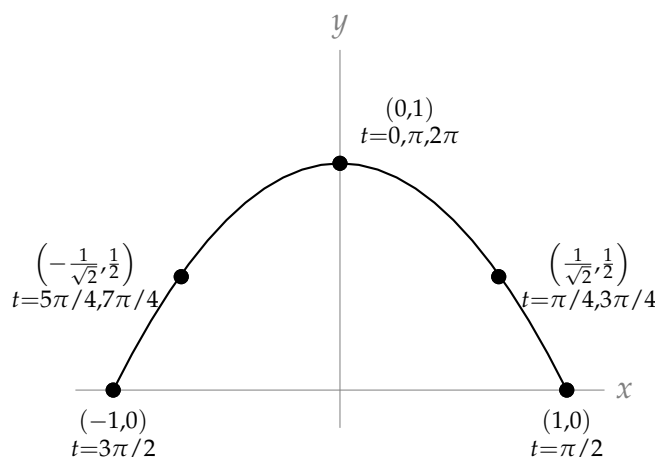
$$\begin{aligned}f'(x) &= \frac{1}{3} \left(\frac{\sqrt{x^3 - 9} \tan x}{e^{\csc x^2}} \right)^{\frac{2}{3}} \left(\frac{\sqrt{x^3 - 9} \tan x \frac{d}{dx} \left\{ e^{\csc x^2} \right\} - e^{\csc x^2} \frac{d}{dx} \left\{ \sqrt{x^3 - 9} \tan x \right\}}{(\tan^2 x)(x^3 - 9)} \right) \\ &= \frac{1}{3} \left(\frac{\sqrt{x^3 - 9} \tan x}{e^{\csc x^2}} \right)^{\frac{2}{3}} \cdot \left(\frac{\sqrt{x^3 - 9} \tan x (-2x) e^{\csc x^2} \csc(x^2) \cot(x^2) - e^{\csc x^2} \left(\frac{3x^2 \tan x}{2\sqrt{x^3 - 9}} + \sqrt{x^3 - 9} \sec^2 x \right)}{(\tan^2 x)(x^3 - 9)} \right)\end{aligned}$$

S-35:

(a) The table below gives us a number of points on our graph, and the times they occur.

t	$(\sin t, \cos^2 t)$
0	(0, 1)
$\pi/4$	$(\frac{1}{\sqrt{2}}, \frac{1}{2})$
$\pi/2$	(1, 0)
$3\pi/4$	$(\frac{1}{\sqrt{2}}, \frac{1}{2})$
π	(0, 1)
$5\pi/4$	$(-\frac{1}{\sqrt{2}}, \frac{1}{2})$
$3\pi/2$	(-1, 0)
$7\pi/4$	$(-\frac{1}{\sqrt{2}}, \frac{1}{2})$
2π	(0, 1)

These points will repeat with a period of 2π . With this information, we have a pretty good idea of the particle's motion:



The particle traces out an arc, pointing down. It starts at $t = 0$ at the top part of the graph at $(1, 0)$, then it moves to the right until it hits $(1, 0)$ at time $t = \pi/2$. From there it reverses direction and moves along the curve to the left, hitting the top at time $t = \pi$ and reaching $(-1, 0)$ at time $t = 3\pi/2$. Then it returns to the top at $t = 2\pi$ and starts again.

So, it starts at the top of the curve, then moves back and forth along the length of the curve. It goes right first, and repeats its cycle every 2π units of time.

(b) Let $y = f(x)$ be the curve the particle traces in the xy -plane. Since x is a function of t , $y(t) = f(x(t))$. What we want to find is $\frac{df}{dx}$ when $t = \left(\frac{10\pi}{3}\right)$. Since $\frac{df}{dx}$ is a function of x , we note that when $t = \left(\frac{10\pi}{3}\right)$, $x = \sin\left(\frac{10\pi}{3}\right) = \sin\left(\frac{4\pi}{3}\right) = -\frac{\sqrt{3}}{2}$. So, the quantity we want to find (the slope of the tangent line to the curve $y = f(x)$ traced by the particle at the time $t = \left(\frac{10\pi}{3}\right)$ is given by $\frac{df}{dx}\left(-\frac{\sqrt{3}}{2}\right)$.

Using the chain rule:

$$\begin{aligned} y(t) &= f(x(t)) \\ \frac{dy}{dt} &= \frac{d}{dt} \{f(x(t))\} = \frac{df}{dx} \cdot \frac{dx}{dt} \\ \text{so, } \frac{df}{dx} &= \frac{dy}{dt} \div \frac{dx}{dt} \end{aligned}$$

Using $y(t) = \cos^2 t$ and $x(t) = \sin t$:

$$\frac{df}{dx} = (-2 \cos t \sin t) \div (\cos t) = -2 \sin t = -2x$$

So, when $t = \frac{10\pi}{3}$ and $x = -\frac{\sqrt{3}}{2}$,

$$\frac{df}{dx}\left(-\frac{\sqrt{3}}{2}\right) = -2 \cdot \frac{-\sqrt{3}}{2} = \sqrt{3}.$$